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SuperMimic: Scaling Humanoid Whole-Body Control with One Differentiable World Model

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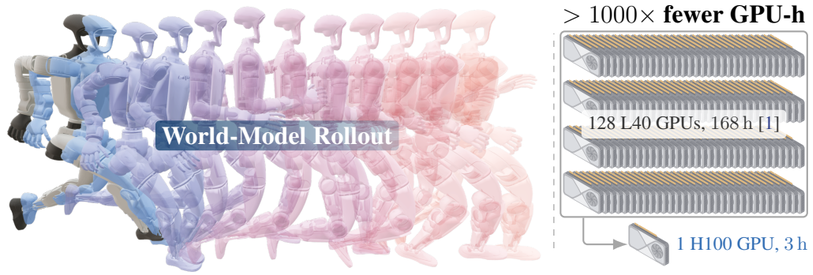


Figure 1: **A learned differentiable world model trains a humanoid whole-body controller and reduces GPU-hours by more than** $1000\times$ **versus cluster-scale model-free reinforcement learning (§4.1).** Left: the policy improves through imagined rollouts of the world model. Right: 128 L40 GPUs train for 168 h in SONIC [1], against one H100 for 3 h in this work.

Abstract: Tracking large human-motion datasets has become the dominant route to general, expressive humanoid whole-body control, but scaling across many motions is expensive. Current pipelines use model-free reinforcement learning, either merging per-motion specialists through lossy distillation, or brute-forcing a single policy with cluster-scale compute. SuperMimic amortizes learning across all motions with one differentiable world model of their shared dynamics, fit from logged transitions of a standard rigid-body simulator. The world model and the tracking policy train together in one loop, and backpropagation-through-time over short imagined rollouts gives the policy analytic gradients. Fast imagined rollouts and analytic gradients make policy learning efficient. In 3 hours on a single GPU, SuperMimic surpasses the return of a state-of-the-art motion tracker baseline trained for 7 days on 128 GPUs. It reduces GPU-hours by more than $1000\times$ and fails on only 25 of $129{,}514$ benchmark motions, versus 179 failures for the baseline. After 9.8 hours on two GPUs, SuperMimic achieves zero failures. From 1 to 264 hours of motion, the world model's prediction error stays stable, and GPU-hours to a target return grow slowly. Our policy transfers to a physical Unitree G1 across diverse and acrobatic motions. SuperMimic removes compute as the bottleneck to scaling humanoid whole-body control, making it practical to measure motion-tracking scaling laws.

Keywords: Humanoid Robot, Whole-Body Control, Model-Based Reinforcement Learning

1 Introduction

Humanoid robots hold the promise of operating in environments designed for people, but doing so requires whole-body controllers that span diverse human motion, from locomotion and dance to crouching and balance recovery. Physics-based motion imitation addresses this by training RL

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SuperMimic：基于单一可微世界模型扩展人形机器人全身控制

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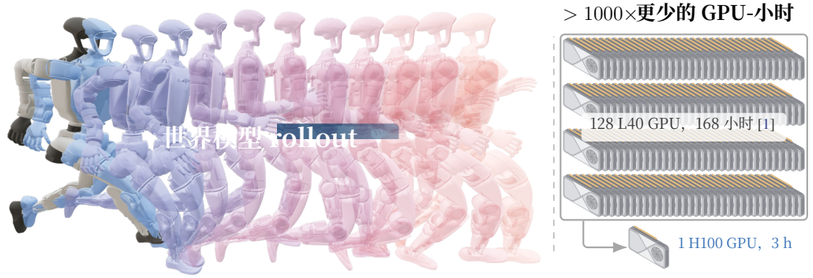


图 1：**一个学习到的可微世界模型训练人形全身控制器，并将 GPU 小时减少超过** $1000\times$ **，相较于集群规模的无模型强化学习（§4.1）**。左图：策略通过世界模型的想象 rollout 得到提升。右图：在 SONIC 中，128 L40 GPU 训练了 168 h [1],，而本工作中仅需一台 H100 训练 3h。

摘要：跟踪大规模人体运动数据集已成为实现通用、高表现力人形全身控制的主要途径，但在多种动作上的扩展成本高昂。现有流程采用无模型强化学习，要么通过有损蒸馏合并针对单一动作的专家策略，要么利用集群规模的算力暴力训练单一策略。SuperMimic 利用一个可微世界模型对所有动作的学习进行摊销，该模型基于标准刚体模拟器记录的转移数据拟合其共享动力学。世界模型与跟踪策略在一个循环中共同训练，通过对短想象 rollout 进行时间反向传播，为策略提供解析梯度。快速的想象 rollout 和解析梯度使得策略学习高效。在单块 GPU 上运行 3 小时，SuperMimic 超越了在 128 块 GPU 上训练 7 天的最先进动作追踪基线的回报。它将 GPU 小时减少了超过 $1000\times$ ，并且在 129,514 个基准动作中仅失败 25 次，而基线失败了 179 次。在两块 GPU 上运行 9.8 小时后，SuperMimic 实现了零失败。从 1 到 264 小时的动作数据，世界模型的预测误差保持稳定，达到目标回报所需的 GPU 小时增长缓慢。我们的策略成功迁移到物理 Unitree G1 机器人上，能够执行多样且杂技般的动作。SuperMimic 消除了计算资源作为扩展人形全身控制的瓶颈，使得测量动作跟踪缩放定律变得切实可行。

关键词：人形机器人、全身控制、基于模型的强化学习

1 引言

人形机器人有望在为人设计的环境中运行，但这需要能够涵盖从运动到舞蹈等各种人类动作的全身控制器。
至蹲伏和平衡恢复。基于物理的动作模仿通过训练强化学习来解决这一问题，

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policies to track reference motion clips [2] drawn from large motion-capture (mocap) datasets [1]. Existing systems scale this idea either by distilling many clip-specific experts into one controller [3] or by training a generalist policy over a large motion library [1]. In both cases, a more diverse motion library produces a more capable controller. But both also become computationally expensive as motion coverage grows: the state-of-the-art tracker SONIC [1], for example, trains for $\sim$168 h on 128 GPUs, totaling roughly 21,000 GPU-hours. Although reference motions differ, they share the same humanoid hybrid rigid-body dynamics. Model-free RL fails to explicitly model this shared structure, instead repeatedly estimating action consequences from noisy simulator rollouts. Scaling to more motions therefore amounts to scaling up compute.

SuperMimic addresses this inefficiency with a single learned, differentiable world model fit from logged simulator transitions. Backpropagation-through-time (BPTT) through short imagined rollouts provides analytic policy gradients through the learned model, allowing one dynamics model to amortize learning across the entire mocap library. Because these gradients come from the learned model rather than from differentiating the simulator, SuperMimic remains compatible with rigid-body simulators, including those with non-smooth contacts and non-differentiable implementations.

On one GPU, SuperMimic surpasses the normalized return of SONIC's released checkpoint within 3 h and reduces GPU-hours by more than 1000$\times$ (§4.1). On a 129,514-motion benchmark, SuperMimic fails on 25 motions after 3 h on one GPU and on none after 9.8 h on two GPUs, against 179 for SONIC's released checkpoint (Appendix F). Across a 264$\times$ increase in dataset length, the world model's prediction error stays stable and SuperMimic's compute to a fixed return grows slowly. The learned policy transfers to a physical Unitree G1. In a preliminary study of downstream interfaces, an RL-trained velocity encoder outperforms a distillation baseline.

Contributions.

- **Model-based RL (MBRL) for deployable humanoid control**. We demonstrate that BPTT through a learned, differentiable world model can train whole-body tracking policies in a setting long dominated by model-free proximal policy optimization (PPO) [4], and that the resulting controller transfers to a physical Unitree G1.
- **Compute is no longer the bottleneck to scaling.** SuperMimic surpasses the return of SONIC, a cluster-scale PPO pipeline, on a single GPU and reduces GPU-hours by more than 1000$\times$ (§4.1). Its training cost grows slowly as the motion library expands, making scaling humanoid whole-body control practical without huge GPU infrastructure.
- **An analysis of the ingredients behind the gains.** We identify three world-model properties behind the compute reduction: prediction error that stays stable over a 264$\times$ increase in dataset length, rollouts more than 10$\times$ faster than the simulator, and analytic gradients, with which the policy reaches the best return of PPO on the world model with about 64$\times$ fewer imagined steps.
- **RL-trained downstream control beyond distillation.** As a preliminary study, the same learned world model trains a downstream goal encoder on the interface's own task reward. On velocity commands, the policy tracks more accurately with this encoder than with a distillation baseline.

2 Related Work

Large-scale physics-based motion tracking. Physics-based motion imitation learns controllers that reproduce reference motions under contact-rich rigid-body dynamics. DeepMimic [2] demonstrated expressive character skills through model-free reinforcement learning, and PHC [11] extended this paradigm to large motion sets with robust recovery. SONIC [1] further demonstrated that scaling motion data, policy capacity, and training compute produces generalist controllers that transfer to humanoid hardware. These results establish motion tracking as a practical route to broad whole-body behavior coverage. However, training these large-scale motion trackers requires extensive simulator rollouts and substantial computation. SuperMimic addresses this bottleneck by learning a differentiable dynamics model shared across all motions, so policy improvement can exploit analytic gradients rather than model-free rollout estimates.

策略以追踪来自大型动作捕捉（mocap）数据集的参考动作片段 [2] [1]。现有系统要么通过将许多片段特定专家蒸馏为一个控制器来扩展这一思路，[3] 要么通过在大型动作库上训练通用策略来实现 [1]。在这两种情况下，更多样化的动作库会产生能力更强的控制器。但随着动作覆盖范围的扩大，两者也都变得计算成本高昂：例如，最先进的追踪器 SONIC [1], 的训练时间为 $\sim$168h，使用 128 个 GPU，总计约21,000 GPU 小时。尽管参考运动各不相同，但它们共享相同的人形混合刚体动力学。无模型强化学习未能显式建模这种共享结构，而是反复从噪声模拟器 rollout 中估计动作后果。因此，扩展到更多动作就等于增加计算量。

SuperMimic 通过一个从记录的模拟器转移中拟合出的单一可微世界模型，解决了这一低效问题。通过对短程想象 rollout 进行时间反向传播（BPTT），该学习模型可提供解析策略梯度，从而使一个动力学模型能够分摊整个动捕库的学习成本。由于这些梯度源自学习模型而非对模拟器的微分，SuperMimic 仍与刚体模拟器兼容，包括那些具有非光滑接触和不可微实现的模拟器。

在单个 GPU 上，SuperMimic 在3 小时内便超越了 SONIC 已发布检查点的归一化回报，并将 GPU 小时减少了超过 1000$\times$ （§4.1）。在一个包含129,514个动作的基准测试中，SuperMimic 在单个 GPU 上运行3 小时后仅在 25 个动作上失败，而在两个 GPU 上运行9.8 小时后无一失败，相比之下 SONIC 已发布的检查点失败了 179 个（附录F）。随着数据集长度增加 264×倍，世界模型的预测误差保持稳定，而 SuperMimic 达到固定回报所需的计算量增长缓慢。所学策略已成功迁移至物理 Unitree G1 机器人。在对下游接口的初步研究中，强化学习训练的速度编码器表现优于蒸馏基线。

贡献。

- **用于可部署人形控制的基于模型的强化学习（MBRL）**。我们证明，通过学习的可微世界模型进行 BPTT，可以在长期由无模型近端策略优化（PPO） [4], 主导的场景中训练全身追踪策略，并且所得控制器可迁移至物理 Unitree G1。
- **算力不再是扩展的瓶颈**。SuperMimic 在单块 GPU 上超越了 SONIC（一个集群规模的 PPO 流水线）的回报，并将 GPU 小时减少了超过 1000$\times$ （§4.1）。
  其训练成本随着动作库的扩展而缓慢增长，使得在不依赖庞大 GPU 基础设施的情况下扩展人形机器人全身控制成为可能。
- **对增益背后关键要素的分析**。我们确定了导致算力减少的三个世界模型属性：预测误差在数据集长度增加时保持稳定、
  rollout 速度比模拟器快超过 10$\times$ 倍，以及解析梯度；借助解析梯度，策略在世界模型上仅需约 64$\times$ 更少的想象步数即可达到 PPO 的最佳回报。
- **超越蒸馏的强化学习训练下游控制**。作为一项初步研究，相同的学习得到的世界模型在接口自身的任务奖励上训练了一个下游目标编码器。在速度指令上，使用该编码器的策略比使用蒸馏基线的策略跟踪得更准确。

2 相关工作

大规模基于物理的动作跟踪。基于物理的动作模仿学习控制器，以在接触丰富的刚体动力学下复现参考动作。DeepMimic [2]通过无模型强化学习展示了富有表现力的角色技能，而 PHC [11] 将这一范式扩展至大型动作集并实现了稳健的恢复能力。SONIC [1] 进一步证明，扩展动作数据、策略容量和训练算力可产生能迁移到人形机器人硬件的通用控制器。这些成果确立了动作跟踪作为实现广泛全身行为覆盖的实用途径。然而，训练这些大规模动作追踪器需要大量的模拟器 rollout 和可观的计算资源。SuperMimic 通过学习一个对所有动作共享的可微动力学模型来解决这一瓶颈，从而使策略改进能够利用解析梯度，而非依赖无模型的 rollout 估计。

Method	Mimic	Latent	Downstream	Task signal	Real robot
NPMP [5]	SVG(0)	BC	SVG(0)	RL	✗
ControlVAE [6]	MBRL		MBRL	RL	✗
PhysicsVAE [7]	PPO	BC[1]	PPO	RL	✗
PULSE [8]	PPO	DAgger	PPO	RL	✗
MaskedMimic [9]	A2C	DAgger		Distill	✗
BeyondMimic [3]	PPO	DAgger	CGD	Distill	✓
SONIC [1]	PPO			Distill[2]	✓
LATENT [10]	PPO	DAgger	PPO	RL	✓
SuperMimic	MBRL			Distill / RL	✓

Table 1: **Comparison of various methods and their training strategies across different stages.** The Mimic, Latent, and Downstream columns indicate the training strategy used at each stage. Task signal is what trains the downstream interface: its own task reward (RL) or a reference-tracking signal (Distill). Merged cells denote jointly trained stages, while adjacent unmerged cells indicate sequential training, with each stage trained after freezing the previous one.
SVG(0): stochastic value gradients; BC: offline behavior cloning on fixed expert rollouts, as opposed to DAgger's student rollouts; CGD: classifier guidance diffusion; A2C: advantage actor–critic; DAgger: dataset aggregation. [1]Behavior cloning with a one-step next-state consistency term through a frozen, learned dynamics model. [2]Goal encoders train jointly with the tracker on the motion-tracking reward and latent-alignment losses; no task-specific reward.

Learned differentiable dynamics for motion control. SuperTrack [12] optimizes motion-control policies through a learned differentiable surrogate of the simulator. ControlVAE [6] jointly learns a dynamics model, latent skill representation, and generative controller, while MoConVQ [13] extends learned-model training to discrete motion representations over tens of hours of data. SuperDyno [14] scales differentiable neural dynamics to tens of thousands of motion clips. These methods establish the effectiveness of learned dynamics and analytic policy gradients for scalable simulated-character control. SuperMimic extends this direction to single-GPU training of a large-scale humanoid tracker and to its sim-to-real deployment: its world model conditions on simulator context, while the deployed actor uses noisy proprioceptive observations and a goal input (§3.2). We evaluate how model accuracy, rollout throughput, and differentiable policy updates contribute to training efficiency, and how prediction error behaves as motion coverage grows (§4.1–4.2), and demonstrate physical deployment of the resulting controller (§4.4).

Model-based policy optimization and robot deployment. Dreamer [15] and DreamerV3 [16] learn latent dynamics for imagination-based actor–critic learning, while TD-MPC [17] combines short-horizon planning with terminal-value prediction. DayDreamer [18] demonstrates world-model learning directly on physical robots. SHAC [19] uses differentiable simulation for short-horizon actor–critic updates, motivating SuperMimic's critic-bootstrapped objective. Analytic-gradient robot control can also combine a simplified differentiable surrogate with a high-fidelity forward simulator, as demonstrated for quadruped locomotion [20]. SuperMimic learns its surrogate from simulator transitions and backpropagates through short imagined rollouts, retaining compatibility with standard rigid-body simulators.

RWM [21] and RWM-U [22] demonstrate that policies trained in learned neural simulators can transfer to physical robots. Both optimize policies using PPO on imagined rollouts; RWM-U additionally uses uncertainty-penalized offline optimization. SuperMimic instead optimizes the policy by differentiating the return through its learned dynamics. RWM and RWM-U pretrain the world model on trajectories from other policies. In contrast, SuperMimic fits its world model during training on data its own policy collects.

Reusable motion representations and downstream control. Reusable controllers connect motion tracking to downstream goal interfaces (Table 1). NPMP [5] and PhysicsVAE [7] learn latent motor representations from expert behavior; PULSE [8] and LATENT [10] use learned motion representations for downstream reinforcement learning. MaskedMimic [9] supports partial-body and multimodal goals through motion inpainting, while BeyondMimic [3] uses guided diffusion for goal

方法	Mimic	隐变量	下游	任务信号	真实机器人
NPMP [5]	SVG(0)	BC	SVG(0)	RL	✗
ControlVAE [6]	MBRL		MBRL	RL	✗
PhysicsVAE [7]	PPO	BC[1]	PPO	RL	✗
PULSE [8]	PPO	DAgger	PPO	RL	✗
MaskedMimic [9]	A2C	DAgger		蒸馏	✗
BeyondMimic [3]	PPO	DAgger	CGD	蒸馏	✓
SONIC [1]	PPO			蒸馏[2]	✓
隐变量 [10]	PPO	DAgger	PPO	RL	✓
SuperMimic	MBRL			蒸馏 / 强化学习	✓

表 1：**不同阶段下各种方法及其训练策略的比较**。Mimic、Latent 和 Downstream 列表示各阶段所使用的训练策略。任务信号用于训练下游接口：可以是其自身的任务奖励（强化学习），也可以是参考跟踪信号（蒸馏）。合并的单元格表示联合训练的阶段，而相邻未合并的单元格表示顺序训练，即每个阶段在前一阶段冻结后进行训练。
SVG(0)：随机值梯度；BC：在固定专家 rollout 上进行的离线行为克隆，与 DAgger 的学生 rollout 相对；CGD：分类器引导扩散；A2C：优势演员 - 评论家；DAgger：数据集聚合。 [1]通过一步下一状态一致性项进行的行为克隆，经由冻结的学习动力学模型。 [2]目标编码器与跟踪器在运动跟踪奖励和隐变量对齐损失上联合训练；不使用特定任务的奖励。

用于运动控制的学习可微动力学。SuperTrack [12]通过模拟器的学习可微代理模型优化运动控制策略。ControlVAE [6]联合学习动力学模型、隐变量技能表示和生成式控制器，而 MoConVQ [13] 将学习模型的训练扩展至涵盖数十小时数据的离散运动表示。SuperDyno [14]将可微神经动力学扩展至数万个运动片段。这些方法确立了学习动力学和解析策略梯度在可扩展仿真角色控制中的有效性。SuperMimic 将这一方向扩展至单 GPU 训练大规模人形机器人跟踪器及其仿真到现实部署：其世界模型以模拟器上下文为条件，而部署的执行器则使用含噪声的本体感觉观测和目标输入（§3.2）。我们评估了模型精度、rollout 吞吐量和可微策略更新对训练效率的贡献，以及预测误差如何随运动覆盖范围增长而变化（§4.1–4.2），并展示了所得控制器的物理部署（§4.4）。

基于模型的策略优化与机器人部署。Dreamer [15] 和 DreamerV3 [16]学习隐变量动力学以进行基于想象的演员 - 评论家学习，而 TD-MPC [17] 则将短视界规划与终端价值预测相结合。DayDreamer [18] 展示了直接在物理机器人上进行世界模型学习。SHAC [19] 利用可微仿真器进行短视界演员 - 评论家更新，从而启发了 SuperMimic 的评论家自举目标。解析梯度机器人控制还可以将简化的可微代理模型与高保真前向仿真器相结合，正如在四足 locomotion [20]中所展示的那样。SuperMimic 从仿真器转移中学习其代理模型，并通过短想象 rollout 进行反向传播，同时保持与标准刚体模拟器的兼容性。

RWM [21] 和 RWM-U [22] 表明，在学习到的神经仿真器中训练的策略可以迁移到物理机器人上。两者均使用 PPO 在想象 rollout 上优化策略；RWM-U 还采用了不确定性惩罚离线优化。相比之下，SuperMimic 通过对其学习动力学的回报进行微分来优化策略。RWM 和 RWM-U 在其他策略产生的轨迹上预训练世界模型。而 SuperMimic 则在其自身策略收集的数据上，于训练过程中拟合其世界模型。

可重用的运动表示与下游控制。可重用控制器将运动跟踪连接至下游目标接口（表1）。NPMP [5] 和 PhysicsVAE [7] 从专家行为中学习隐变量运动表示；PULSE [8] 和 LATENT [10] 利用学习到的运动表示进行下游强化学习。MaskedMimic [9] 通过运动修复支持部分身体和多模态目标，而 BeyondMimic [3]则利用引导扩散实现目标

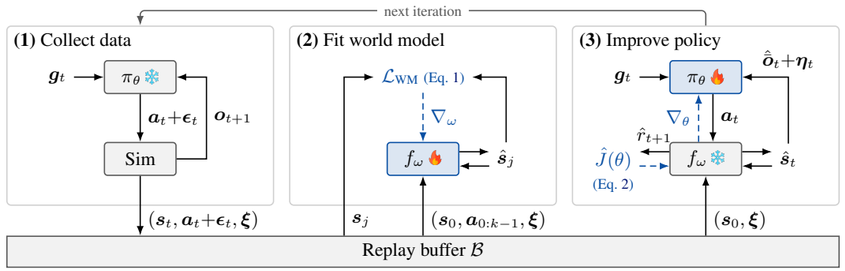


Figure 2: **SuperMimic training loop.** (1) The current policy π_θ collects transitions in simulation into a fixed-size replay buffer $\mathcal{B}$, where new transitions replace the oldest. (2) The differentiable world model is fit to k-step windows sampled from $\mathcal{B}$ (§3.2). (3) The policy is improved by backpropagation-through-time through the frozen world model over H imagined steps; a critic V_ψ, trained in the same phase, bootstraps the return beyond H (§3.3). Dashed arrows are gradients; ❄ = frozen, 🔥 = trained. Symbols: goal $\boldsymbol{g}_t$ (e.g., reference motion), action $\boldsymbol{a}_t$, action noise $\boldsymbol{\epsilon}_t$, observation $\boldsymbol{o}_t$, state $\boldsymbol{s}$ and predicted state $\hat{\boldsymbol{s}}$, and sim context $\boldsymbol{\xi}$ (domain-randomization parameters). $\mathcal{L}_{\text{WM}}$ is the world-model loss, $\hat{\boldsymbol{o}}_t$ the predicted noise-free observation, $\boldsymbol{\eta}_t$ the sensor noise, $\hat{r}_{t+1}$ the predicted reward, and $\hat{J}(\theta)$ the policy objective.

following. SONIC [1] jointly trains goal encoders with its tracker using tracking rewards and latent-alignment losses. ControlVAE [6] optimizes downstream task policies through its learned model while freezing the skill controller. SuperMimic jointly trains goal-specific encoders and a shared decoder through the same dynamics model (Appendix B). Goal interfaces can optimize either a reference-tracking reward or their own task reward. Our preliminary study trains a velocity interface concurrently with the motion-tracking interface, once on each objective (§4.5).

3 Method

3.1 SuperMimic training loop

SuperMimic trains one goal-conditioned whole-body controller across a multi-skill motion library and supports multiple goal interfaces. It captures the shared humanoid dynamics in a differentiable world model and improves the policy through short imagined rollouts. Training alternates between three phases in one loop (Figure 2). First, the current policy collects simulator transitions into a replay buffer. Second, we fit the differentiable world model to short replay windows. Third, the policy is improved by backpropagating an imagined return through the frozen world model, with each rollout seeded from a replay-buffer state in the Dyna style [23, 24]. The buffer, model, and policy are updated throughout training, so dynamics learning and control improve together rather than in separate stages (Algorithm A.1, Appendix D). Transitions from earlier policies remain valid for fitting the world model, because the dynamics do not depend on the policy that collected them.

We denote the goal-conditioned partially observable Markov decision process (POMDP) by $\mathcal{M}$ and use compact notation with state $\boldsymbol{s}_t$, goal $\boldsymbol{g}_t$, and action $\boldsymbol{a}_t$. State $\boldsymbol{s}_t$ gives the proprioceptive observation $\boldsymbol{o}_t$ and the termination d_t, which is 1 if $\boldsymbol{s}_t$ is terminal. Action $\boldsymbol{a}_t$ in state $\boldsymbol{s}_t$ yields the reward r_{t+1} and the next state $\boldsymbol{s}_{t+1}$. The policy is deterministic, $\boldsymbol{a}_t = \pi_\theta(\boldsymbol{o}_t, \boldsymbol{g}_t)$, and $\boldsymbol{a}_t$ parameterizes a normalized joint setpoint; for motion tracking, $\boldsymbol{g}_t$ is read from a reference motion. During data collection, the simulator executes the noisy action $\boldsymbol{a}_t + \boldsymbol{\epsilon}_t$ with action noise $\boldsymbol{\epsilon}_t \sim \rho_\epsilon$, and the replay buffer stores this executed noisy action. The noise widens the range of actions on which the world model is fit; it serves only the model fit and is optional. Returns use a soft survival weight, $c_0 = 1$ and $c_{t+1} = c_t(1 - d_{t+1})$, giving $J(\theta) = \mathbb{E}[\sum_{t\geq 0} \gamma^t c_t r_{t+1}]$ with discount $\gamma \in [0, 1)$. The simulator logs hard terminations, while the learned model predicts a termination probability $\hat{d}_t \in [0, 1]$, which keeps the survival weight differentiable in imagined rollouts. Throughout, a hat marks the world-model counterpart of a simulator quantity. We also factor the observation as

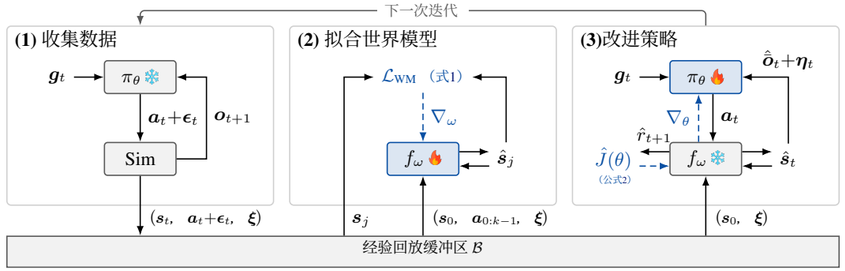


图 2：SuperMimic 训练循环。(1) 当前策略 π_θ 在仿真中收集转移数据并存入固定大小的经验回放缓冲区 $\mathcal{B}$，新转移数据将替换最旧的数据。(2) 可微分世界模型拟合从 $\mathcal{B}$ 中采样的 k步窗口（§3.2）。(3) 通过在冻结的世界模型上对 H 个想象步数进行时间反向传播来改进策略；critic V_ψ在同一阶段训练，用于对超出 H 的回报进行自举（§3.3）。虚线箭头表示梯度；= 为❄冻结状态，=为🔥训练状态。符号说明：目标 $\boldsymbol{g}_t$（例如参考动作）、动作 $\boldsymbol{a}_t$、动作噪声 $\boldsymbol{\epsilon}_t$、观测 $\boldsymbol{o}_t$、状态 $\boldsymbol{s}$ 与预测状态$\hat{\boldsymbol{s}}$，以及仿真上下文 $\boldsymbol{\xi}$（域随机化参数）。$\mathcal{L}_{\text{WM}}$ 是世界模型损失，$\hat{\boldsymbol{o}}_t$ 是预测的无噪声观测，$\boldsymbol{\eta}_t$ 是传感器噪声，$\hat{r}_{t+1}$ 是预测奖励，$J(\theta)$ 是策略目标。

跟随。SONIC [1] 利用跟踪奖励和隐变量对齐损失，联合训练其跟踪器的目标编码器。ControlVAE [6] 在冻结技能控制器的同时，通过其学习到的模型优化下游任务策略。SuperMimic 通过同一动力学模型联合训练目标特定编码器和共享解码器（附录B）。目标接口可以优化参考跟踪奖励或其自身的任务奖励。我们的初步研究将速度接口与运动追踪接口并发训练，每个目标各训练一次（§4.5）。

3 方法

3.1 SuperMimic 训练循环

SuperMimic 在一个多技能动作库上训练一个目标条件化全身控制器，并支持多种目标接口。它在可微世界模型中捕捉共享人形动力学，并通过短想象 rollout 改进策略。训练在一个循环中交替进行三个阶段（图2）。首先，当前策略将仿真器转移收集到经验回放缓冲区中。其次，我们将可微世界模型拟合到短回放窗口。第三，通过在冻结的世界模型中反向传播想象回报来改进策略，每个 rollout 均以 Dyna 风格从回放缓冲区状态 seeding。缓冲区、模型和策略在整个训练过程中不断更新，因此动力学学习与控制协同改进，而非分阶段进行（算法A.1，附录D）。来自早期策略的转移对于拟合世界模型仍然有效，因为动力学不依赖于收集它们的策略。

我们将目标条件化部分可观测马尔可夫决策过程（POMDP）表示为 $\mathcal{M}$，并使用紧凑符号，其中状态为 $\boldsymbol{s}_t$，目标为 $\boldsymbol{g}_t$，动作为 $\boldsymbol{a}_t$。状态 $\boldsymbol{s}_t$ 提供本体感觉观测 $\boldsymbol{o}_t$ 和终止条件 d_t，若 $\boldsymbol{s}_t$ 处于终端状态，则其值为 1。在状态 $\boldsymbol{s}_t$ 下执行动作 $\boldsymbol{a}_t$ 会产生奖励 r_{t+1} 和下一个状态 $\boldsymbol{s}_{t+1}$。策略是确定性的，$\boldsymbol{a}_t = \pi_\theta(\boldsymbol{o}_t, \boldsymbol{g}_t)$，且 $\boldsymbol{a}_t$ 参数化一个归一化关节设定点；对于动作跟踪，$\boldsymbol{g}_t$ 从参考动作中读取。在数据收集过程中，模拟器执行带有动作噪声 $\boldsymbol{\epsilon}_t \sim \rho_\epsilon$ 的含噪动作 $\boldsymbol{a}_t + \boldsymbol{\epsilon}_t$，经验回放缓冲区会存储该执行的含噪动作。噪声扩大了世界模型拟合所依据的动作范围；它仅服务于模型拟合，且是可选的。回报使用软生存权重 $c_0 = 1$ 和 $c_{t+1} = c_t(1 - d_{t+1})$，从而得到具有折扣 $\gamma \in [0, 1)$ 的 $J(\theta) = \mathbb{E}[\sum_{t\geq 0} \gamma^t c_t r_{t+1}]$。模拟器记录硬终止，而学习到的模型预测终止概率 $\hat{d}_t \in [0, 1]$，这使得生存权重在想象 rollout 中保持可微。全文中，帽子符号标记模拟器量的世界模型对应量。我们还将观测分解为

$\boldsymbol{o}_t = \bar{\boldsymbol{o}}_t + \boldsymbol{\eta}_t$, with $\bar{\boldsymbol{o}}_t$ the noise-free observation and known observation noise $\boldsymbol{\eta}_t \sim \rho_\eta$ sampled outside the world model. Appendix A gives the full spaces and reward.

The default interface is full-reference motion tracking. Other goal interfaces, such as velocity, share the same learned dynamics and train jointly with motion tracking (Appendix B).

3.2 Learning a differentiable world model

The simulator's articulated multi-rigid-body dynamics are known, but the simulator need not provide gradients, so SuperMimic learns a differentiable surrogate to obtain policy gradients: it supplies the predictions and Jacobians for the BPTT update in §3.3. The resulting analytic policy gradient is the gradient of a surrogate for the expected return in simulation. Since the simulator's randomness has known sources, the world model stays deterministic: it conditions on the per-episode sim context $\boldsymbol{\xi}$ (friction, calibration offsets, and torso center-of-mass) and predicts the noise-free observation $\bar{\boldsymbol{o}}_t$.

The world model contains a learned dynamics model and reward, observation, and termination heads; each head is either learned or computed analytically,

$$\hat{\boldsymbol{s}}_{t+1} = f_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{a}_t, \boldsymbol{\xi}), \qquad \hat{r}_{t+1} = r_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{a}_t, \boldsymbol{g}_t), \qquad \hat{\bar{\boldsymbol{o}}}_t = g_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{\xi}), \qquad \hat{d}_t = d_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{g}_t).$$

We fit the network weights ω on k-step autoregressive replay windows (k=8), seeded at the logged state $\hat{\boldsymbol{s}}_0 = \boldsymbol{s}_0$ and rolled forward on logged actions, by minimizing

$$\mathcal{L}_{\text{WM}}(\omega) = \mathbb{E}_{\mathcal{B}} \sum_{j=1}^{k} \Big[\|\hat{\boldsymbol{s}}_j - \boldsymbol{s}_j\|^2 + (\hat{r}_j - r_j)^2 + \|\hat{\bar{\boldsymbol{o}}}_j - \bar{\boldsymbol{o}}_j\|^2 + \text{BCE}(\hat{d}_j, d_j) \Big], \tag{1}$$

where an analytic head has no loss term. Here r_j is the simulator reward, $d_j \in \{0, 1\}$ is the simulator termination label, and BCE is the binary cross-entropy.

3.3 Learning the policy through the world model

During each policy update, we hold the world-model weights ω fixed and optimize π_θ and V_ψ on imagined rollouts of H=32 steps. The critic V_ψ estimates the return beyond step H, so the model must be accurate only within these H steps. This imagination horizon exceeds the k-step replay windows on which we fit the model (§3.2). In practice, the learned model remains stable over indefinitely long rollouts, because the k-step autoregressive fit trains it to correct its own prediction errors.

Each imagined rollout starts from a replay-buffer state $\boldsymbol{s}_0 \sim \rho_0$ with its recorded sim context $\boldsymbol{\xi}$. At step t, the model emits $\hat{\bar{\boldsymbol{o}}}_t$, we re-add sensor noise, and the policy acts: $\boldsymbol{a}_t = \pi_\theta(\hat{\bar{\boldsymbol{o}}}_t + \boldsymbol{\eta}_t, \boldsymbol{g}_t)$, $\boldsymbol{\eta}_t \sim \rho_\eta$. The policy objective over imagined rollouts is

$$\hat{J}(\theta) = \mathbb{E}_{\boldsymbol{s}_0 \sim \rho_0,\, \pi_\theta} \Big[\sum_{t=0}^{H-1} \gamma^t \hat{c}_t\, \hat{r}_{t+1} \;+\; \gamma^H \hat{c}_H\, V_\psi(\hat{\boldsymbol{s}}_H, \boldsymbol{g}_H, \boldsymbol{\xi}) \Big]. \tag{2}$$

For motion tracking, the reward r_{t+1} is the BeyondMimic motion-tracking reward [3] without its self-collision penalty (Appendix A).

The world model is differentiable, so BPTT computes a first-order gradient $\nabla_\theta \hat{J}$ through the H imagined steps via the world model's Jacobians with respect to $\hat{\boldsymbol{s}}_t$ and $\boldsymbol{a}_t$. In contrast, PPO treats the dynamics as a black box: it weights the score $\nabla_\theta \log \pi_\theta$ of a stochastic policy by the advantages of sampled actions, a likelihood-ratio estimate. BPTT instead differentiates the return with respect to each action, so over short horizons its gradient has much lower variance.

The critic V_ψ is fit to generalized advantage estimation returns [25] computed backward in time over the imagined transitions only,

$$\mathcal{L}_V(\psi) = \mathbb{E} \sum_{t=0}^{H-1} \big(V_t - \hat{R}_t\big)^2, \qquad \hat{R}_t = \hat{r}_{t+1} + \gamma(1 - \hat{d}_{t+1}) \big[(1-\lambda) V_{t+1} + \lambda \hat{R}_{t+1}\big], \tag{3}$$

with $V_t = V_\psi(\hat{\boldsymbol{s}}_t, \boldsymbol{g}_t, \boldsymbol{\xi})$. The recursion starts from $\hat{R}_H = V_H$, the critic's estimate of the return beyond the horizon. The targets $\hat{R}_t$ are computed once per rollout and remain fixed while ψ is fit.

$\boldsymbol{o}_t = \bar{\boldsymbol{o}}_t + \boldsymbol{\eta}_t$，其中 $\bar{\boldsymbol{o}}_t$ 为无噪声观测，已知观测噪声 $\boldsymbol{\eta}_t \sim \rho_\eta$ 在世界模型之外采样。附录 A 提供了完整的空间定义和奖励。

默认接口是全参考运动追踪。其他目标接口（如速度）共享相同的学习动力学模型，并与动作跟踪联合训练（附录B）。

3.2 学习可微世界模型

模拟器的关节多刚体动力学是已知的，但模拟器无需提供梯度，因此 SuperMimic 学习一个可微代理模型以获取策略梯度：它为 § 3.3中的 BPTT 更新提供预测和雅可比矩阵。所得的解析策略梯度是仿真中期望回报代理模型的梯度。由于模拟器的随机性来源已知，世界模型保持确定性：它以每集的仿真上下文 $\boldsymbol{\xi}$（摩擦、校准偏移和躯干质心）为条件，并预测无噪声观测 $\bar{\boldsymbol{o}}_t$。

世界模型包含一个学习动力学模型以及奖励、观测和终止头；每个头要么是学习得到的，要么是通过解析计算得出的，

$$\hat{\boldsymbol{s}}_{t+1} = f_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{a}_t, \boldsymbol{\xi}), \qquad \hat{r}_{t+1} = r_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{a}_t, \boldsymbol{g}_t), \qquad \hat{\bar{\boldsymbol{o}}}_t = g_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{\xi}), \qquad \hat{d}_t = d_\omega(\hat{\boldsymbol{s}}_t, \boldsymbol{g}_t).$$

我们通过最小化以下目标，在记录状态 $\hat{\boldsymbol{s}}_0 = \boldsymbol{s}_0$ 处初始化并在记录动作上向前推演的 k步自回归回放窗口（k=8）上拟合网络权重 ω ，

$$\mathcal{L}_{\text{WM}}(\omega) = \mathbb{E}_{\mathcal{B}} \sum_{j=1}^{k} \Big[\|\hat{\boldsymbol{s}}_j - \boldsymbol{s}_j\|^2 + (\hat{r}_j - r_j)^2 + \|\hat{\bar{\boldsymbol{o}}}_j - \bar{\boldsymbol{o}}_j\|^2 + \text{BCE}(\hat{d}_j, d_j) \Big], \tag{1}$$

其中解析头没有损失项。这里 r_j 是模拟器奖励， $d_j \in \{0, 1\}$ 是模拟器终止标签，BCE 是二元交叉熵。

3.3 通过世界模型学习策略

在每次策略更新期间，我们保持世界模型权重 ω 固定，并在 π_θ 步的想象 rollout 上优化 V_ψ 和 H=32 。critic V_ψ 估计超出第 H步的回报，因此模型只需在这些 H 步内保持准确。这一想象范围超过了我们用于拟合模型的 k步回放窗口（ § 3.2）。在实践中，学习到的模型在无限长的 rollout 中保持稳定，因为 k步自回归拟合训练其修正自身的预测误差。

每次想象 rollout 都从回放缓冲区状态 $\boldsymbol{s}_0 \sim \rho_0$ 开始，并携带其记录的仿真上下文 $\boldsymbol{\xi}$。在第 t 步，模型输出 $\hat{\bar{\boldsymbol{o}}}_t$，我们重新添加传感器噪声，然后策略执行动作： $\boldsymbol{a}_t = \pi_\theta(\hat{\bar{\boldsymbol{o}}}_t + \boldsymbol{\eta}_t, \boldsymbol{g}_t), \boldsymbol{\eta}_t \sim \rho_\eta$。基于想象 rollout 的策略目标为

$$\hat{J}(\theta) = \mathbb{E}_{\boldsymbol{s}_0 \sim \rho_0,\, \pi_\theta} \Big[\sum_{t=0}^{H-1} \gamma^t \hat{c}_t\, \hat{r}_{t+1} \;+\; \gamma^H \hat{c}_H\, V_\psi(\hat{\boldsymbol{s}}_H, \boldsymbol{g}_H, \boldsymbol{\xi}) \Big]. \tag{2}$$

对于动作跟踪，奖励 r_{t+1} 是 BeyondMimic 运动追踪奖励 [3]（不含其自碰撞惩罚）（附录A）。

$\hat{}$世界模型是可微的，因此 BPTT 通过世界模型关于 $\hat{}\boldsymbol{s}_t$ 和 $\boldsymbol{a}_t$的雅可比矩阵，计算穿过 H想象步数的一阶梯度 $\nabla_\theta J$ 。相比之下，PPO 将动力学视为黑盒：它用采样动作的优势对随机策略的得分 $\nabla_\theta \log \pi_\theta$ 进行加权，这是一种似然比估计。而 BPTT 直接对每个动作的回报求导，因此在短 horizon 内，其梯度具有更低的方差。

critic V_ψ 被拟合为广义优势估计回报 [25]，该回报仅在想象转移上反向时间计算，

$$\mathcal{L}_V(\psi) = \mathbb{E} \sum_{t=0}^{H-1} \big(V_t - \hat{R}_t\big)^2, \qquad \hat{R}_t = \hat{r}_{t+1} + \gamma(1 - \hat{d}_{t+1}) \big[(1-\lambda) V_{t+1} + \lambda \hat{R}_{t+1}\big], \tag{3}$$

其中包含 $V_t = V_\psi(\hat{\boldsymbol{s}}_t, \boldsymbol{g}_t, \boldsymbol{\xi})$。递归从 $\hat{R}_H = V_H$ 开始，即 critic 对超出 horizon 的回报的估计值$\hat{}$。目标值 R_t 在每个 rollout 中计算一次，并在拟合 ψ 时保持不变。

The policy and critic updates together form a short-horizon actor–critic in the spirit of SHAC [19], which backpropagates through a differentiable simulator. SuperMimic replaces SHAC's hard episode termination with the survival weight $\hat{c}_t$, and its critic has no exponential-moving-average target network. Prior motion-tracking controllers trained with analytic gradients often omit a critic and optimize only the H-step reward sum [12, 6]. Without a critic, training can reach a similar nominal tracking return, but this unbootstrapped objective is more myopic. The critic matters after perturbations, where recovery or balance rewards can lie beyond the imagination window (§4.3).

4 Experimental Results

We show that SuperMimic scales favorably relative to PPO pipelines, identify the world-model properties behind this scaling, demonstrate transfer on a real Unitree G1, and report a preliminary study of downstream goal interfaces. The simulator is MuJoCo Warp, a GPU implementation of MuJoCo [26]. All our reported costs are measured on NVIDIA H100 (80 GB SXM5) GPUs of Lambda on-demand instances. We ran several random seeds and found training very stable, which we attribute to the dense reward that motion tracking gives at every step. The figures therefore do not report seed variation. Normalized return, return for short, is the undiscounted sum of the per-step reward over the full episode, divided by the largest value this sum can take. A policy that terminates early therefore scores low.

4.1 SuperMimic reaches SONIC-level tracking with far fewer GPU-hours

We compare against SONIC, a published PPO pipeline trained on a 128-GPU cluster. Both methods build on the BeyondMimic motion-tracking formulation [3], making SONIC the closest available published comparison. Within 3 h on one GPU, SuperMimic surpasses the normalized return of the released SONIC checkpoint (Figure 3(a)). On a shared benchmark of 129,514 motions, it also fails on fewer motions than SONIC: 25 after 3 h on one GPU and 0 after 9.8 h on two GPUs, against 179 for SONIC's released checkpoint (Appendix F). We state the 3 h conservatively: from then on, every checkpoint is ahead of SONIC by at least half of the gap between SONIC and SuperMimic's best result, in both return and failed motions. SuperMimic thus uses more than $50\times$ less wall-clock time and more than $1000\times$ fewer GPU-hours than SONIC.

Compute accounting. The ratios compare SONIC's reported 128 NVIDIA L40 GPUs for 7 days (about 21,000 GPU-hours) with 3 h on one NVIDIA H100, within which SuperMimic surpasses the return of SONIC's released checkpoint. They are about $56\times$ ($>50\times$) in wall-clock time and about $7200\times$ ($>1000\times$) in GPU-hours. The 3 h cover the full training loop; checkpoint evaluations run after training and are not counted. SuperMimic trains on 264.56 h of motion from BONES-SEED [27], the public subset of SONIC's data, about $2.3\times$ less than SONIC's 611 h. The GPU models also differ: in peak throughput, an H100 delivers $0.74\times$ the fp32 and $5.5\times$ the tensor-core performance of an L40; how much of that peak either run used is unknown. Nonetheless, $2.3\times$ in data and at most $5.5\times$ in peak GPU throughput cannot account for a $1000\times$ gap in GPU-hours.

Figure 4(a) gives a controlled comparison on LAFAN1 [28]. The baseline, PPO-Sim, is PPO trained in the simulator. PPO-Sim shares SuperMimic's motions, observations, reward, termination, domain randomization, actor and critic network sizes, and evaluation protocol. It uses BeyondMimic's PPO hyperparameters [3] and 8192 parallel environments per GPU, twice SuperMimic's. PPO-Sim estimates its policy gradient from sampled simulator rollouts; SuperMimic backpropagates its imagined return through world-model rollouts. SuperMimic reaches a return of 0.82 in evaluation within 1.2 h on one GPU. Within the 18 h budget, PPO-Sim does not reach it. Its final return is 0.60 on one GPU, 0.70 on four, and 0.72 on eight, so going from four to eight GPUs barely helps.

Figure 3(b, c) shows that, for SuperMimic, GPU-hours to the target return grow slowly as dataset length increases, while on the same sweep the world model's prediction error stays stable; §4.2 traces both trends to one world model amortizing the dynamics shared across motions.

策略与价值网络更新共同构成了一种短视界演员 - 评论家算法，其理念源于 SHAC [19]，该算法通过可微分模拟器进行反向传播。SuperMimic 将 SHAC 中的硬性 episode 终止条件替换为生存权重$\hat{c}_t$，且其 critic 不包含指数移动平均目标网络。先前使用解析梯度训练的动作跟踪控制器通常省略 critic，仅优化 H步奖励总和 [12, 6]。若无 critic，训练虽可达到类似的标称跟踪回报，但这种无自举目标更为短视。在受到扰动后，critic 显得尤为重要，因为恢复或平衡相关的奖励可能位于想象窗口之外（§4.3）。

4 实验结果

我们表明，SuperMimic 相较于 PPO 流程具有更优越的扩展性，揭示了促成这种扩展性的世界模型属性，在真实的 Unitree G1 上展示了迁移能力，并报告了关于下游目标接口的初步研究。模拟器采用 MuJoCo Warp，这是 MuJoCo 的 GPU 实现 [26]。我们报告的所有成本均基于 Lambda 按需实例上的 NVIDIA H100（80 GB SXM5）GPU 进行测量。我们运行了多个随机种子，发现训练非常稳定，我们将此归因于动作跟踪在每一步提供的稠密奖励。因此，图表中未报告种子变化。归一化回报（简称回报）是指整个 episode 中每步奖励的无折扣和除以该和可能达到的最大值。因此，提前终止的策略得分较低。

4.1 SuperMimic 以远少于 SONIC 的 GPU 小时达到同等水平的动作跟踪

我们与 SONIC 进行了对比，SONIC 是一个已发布的 PPO 流程，在 128 GPU 集群上训练而成。两种方法均基于 BeyondMimic 动作跟踪公式 [3], 这使得 SONIC 成为最接近的可用公开对比对象。在单块 GPU 上运行3h 后，SuperMimic 的归一化回报便超过了已发布的 SONIC 检查点（图3(a)）。在包含129,514 个动作的共享基准测试中，SuperMimic 失败的动作数量也少于 SONIC：在单块 GPU 上运行3 h 后为 25 个，在两块 GPU 上运行9.8h 后为 0 个，而 SONIC 已发布检查点的失败动作为 179 个（附录F）。我们保守地陈述3h 这一时间点：从此时起，每个检查点在回报和失败动作两方面均领先于 SONIC，且领先幅度至少为 SONIC 与 SuperMimic 最佳结果之间差距的一半。因此，SuperMimic 所用的挂钟时间比 SONIC 少超过 $50\times$ 倍，所用的 GPU 小时比 SONIC 少超过 $1000\times$ 倍。

计算核算。这些比率将 SONIC 报告的 128 块 NVIDIA L40 GPU 运行 7 天（约 21,000 GPU 小时）与在单块 NVIDIA H100 上运行 3 小时进行了比较，在此期间 SuperMimic 的表现超越了 SONIC 已发布检查点的回报。它们在挂钟时间上约为 $56\times$ ($>50\times$)，在 GPU 小时上约为 $7200\times$ ($>1000\times$)。这 3 小时涵盖了完整训练循环；检查点评估在训练之后运行，未计入在内。SuperMimic 使用来自 BONES-SEED 的 264.56 小时动作数据进行训练 [27], SONIC 数据的公开子集，比 SONIC 的 611 小时少了约 $2.3\times$ 。GPU 模型也存在差异：就峰值吞吐量而言，H100 提供的是 L40 的 fp32 性能 $0.74\times$ 和张量核心性能 $5.5\times$ ；但两者实际利用了多少峰值性能尚不清楚。尽管如此，数据方面 $2.3\times$ 以及峰值 GPU 吞吐量方面最多 $5.5\times$ 的差异，都无法解释 GPU 小时上 $1000\times$的差距。

图4(a) 展示了在 LAFAN 上的受控对比1 [28]。基线方法 PPO-Sim 是在模拟器中训练的 PPO。PPO-Sim 与 SuperMimic 共享相同的动作、观测、奖励、终止条件、域随机化、actor 和 critic 网络规模以及评估协议。它采用 BeyondMimic 的 PPO 超参数 [3] ，并且每个 GPU 使用 8192 个并行环境，是 SuperMimic 的两倍。PPO-Sim 通过采样的模拟器 rollout 估计其策略梯度；而 SuperMimic 则通过世界模型 rollout 反向传播其想象回报。SuperMimic 在一块 GPU 上于1.2 小时内即可在评估中达到 0.82 的回报。而在18 小时的预算内，PPO-Sim 未能达到该水平。其最终回报在单块 GPU 上为 0.60，四块 GPU 上为 0.70，八块 GPU 上为 0.72，因此从四块 GPU 增加到八块 GPU 几乎无益。

图3(b, c) 显示，对于 SuperMimic，随着数据集长度增加，达到目标回报所需的 GPU 小时数增长缓慢，而在同一扫描中，世界模型的预测误差保持稳定；§4.2将这两种趋势归因于一个世界模型摊销了跨动作共享的动力学。

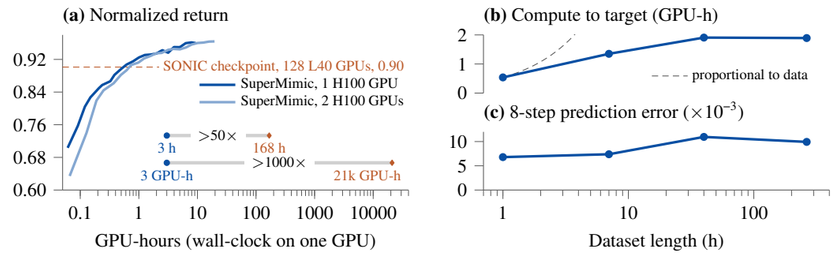


Figure 3: **SuperMimic surpasses SONIC's tracking return with far fewer GPU-hours.** (a) SuperMimic on one and two H100 GPUs surpasses the return of the released SONIC checkpoint (dashed), all three scored on the same benchmark of 129,514 motions (264 h). Two budget rows compare SuperMimic (dot) with SONIC (diamond): SuperMimic reduces wall-clock time by more than 50× and GPU-hours by more than 1000×. (b, c) On a shared dataset-length sweep, GPU-hours to target return (b) grow slowly while the world model's 8-step prediction error (c) stays within 0.007–0.011 as the dataset grows from 1 to 264 h, consistent with one world model amortizing shared body and contact dynamics across motions. The prediction error counts the state only, is normalized by a predefined nominal error, and is measured online on policy rollouts.

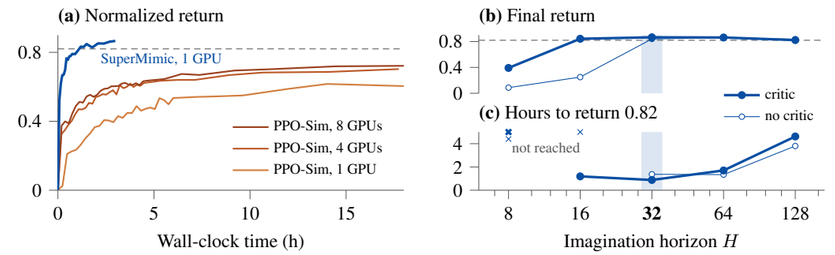


Figure 4: **In controlled comparisons, SuperMimic reaches a higher return sooner than PPO-Sim, and the critic enables shorter imagination horizons.** All runs use LAFAN1 (40 clips, 2.45 h). (a) SuperMimic backpropagates its imagined return through its world model on one GPU; PPO-Sim runs PPO in the simulator on 1, 4, or 8 GPUs (§4.1). Each point on a curve is a saved checkpoint scored on all 40 clips. (b, c) Final training return (b) and wall-clock hours until the training return reaches 0.82 (c) for each imagination horizon H, with and without the critic. The dashed line in (a) and (b) marks this target return of 0.82. Training return is the return of training episodes. Their initial states follow return-based reset sampling (Appendix D.1) and shift during training, so training return differs from the checkpoint scores in (a). In (c), a cross marks a horizon at which return 0.82 is not reached: both variants at H=8, no critic at H=16. The shaded band marks the default H=32.

4.2 Properties of the learned world model

These results rest on three properties of the learned world model f_ω: its accuracy holds as the motion library grows, and its rollouts are fast and differentiable.

One model amortizes the motion library. In Figure 3(c), the world model's 8-step prediction error stays between 0.007 and 0.011 as the dataset grows from 1 to 264 h. Every motion is executed by the same humanoid, so added clips supply more evidence about the shared body and contact dynamics rather than demanding new capacity. This is consistent with the slow growth of the compute needed to reach the target return (§4.1).

Imagination accelerates rollouts. During policy optimization, rollout speed sets the update cost. Figure 5(a) shows that the world model runs more than 10× faster than MuJoCo Warp at the batch sizes used in practice, and up to 62× at 64k parallel environments.

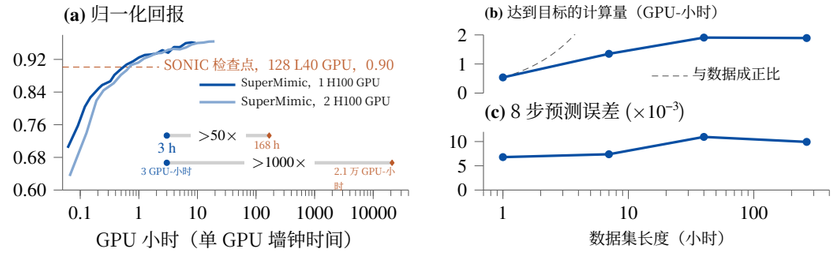


图 3：**SuperMimic 以远更少的 GPU 小时超越了 SONIC 的跟踪回报**。(a) 在一台和两台 H100 GPU 上运行的 SuperMimic 超越了发布的 SONIC 检查点（虚线）的回报，三者均在相同的包含 129,514 个动作（264 小时）的基准测试上进行评分。两行预算数据比较了 SuperMimic（圆点）与 SONIC（菱形）：SuperMimic 将挂钟时间减少了超过 50×，并将 GPU 小时减少了超过 1000×。(b, c) 在共享的数据集长度扫描中，达到目标回报所需的 GPU 小时（b）增长缓慢，而随着数据集从 1 小时增长到 264 小时，世界模型的 8 步预测误差（c）保持在 0.007–0.011 之间，这与一个世界模型在不同动作间分摊共享身体和接触动力学的情况一致。该预测误差仅统计状态，由预定义的标称误差进行归一化，并在策略推出过程中在线测量。

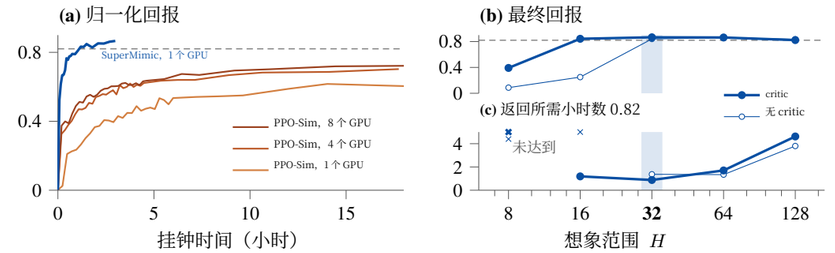


图 4：**在受控对比中，SuperMimic 比 PPO-Sim 更快达到更高的回报，且 critic 能够缩短想象范围**。所有运行均使用 LAFAN1（40 个片段，2.45 小时）。(a) SuperMimic 在其世界模型上通过单块 GPU 反向传播其想象回报；PPO-Sim 则在模拟器中以 1、4 或 8 块 GPU 运行 PPO（§4.1）。曲线上的每个点均为一个保存的检查点，并在全部 40 个片段上进行评分。(b, c) 针对不同想象范围，展示最终训练回报 (b) 以及训练回报达到 0.82 所需的挂钟时间（小时）(c)，分别包含和不包含 critic 的情况。(a) 和 (b) 中的虚线标示了 0.82 这一目标回报值。训练回报是指训练剧集的回报。其初始状态遵循基于回报的复位采样（附录D.1），并在训练过程中发生偏移，因此训练回报不同于 (a) 中的检查点评分。在 (c) 中，叉号标记了未能达到 0.82 回报值的想象范围：两种变体均在 处，无 critic 变体在 处。阴影带标示了默认的 。

4.2 所学世界模型的特性

这些结果依赖于学习得到的世界模型 f_ω 的三个特性：其准确性随着动作库的增长而保持，且其 rollout 快速且可微。

一个模型摊薄了动作库的成本。在图3(c) 中，随着数据集从 1 小时增长到264h，世界模型的 8 步预测误差保持在 0.007 到 0.011 之间。每个动作均由同一个人形机器人执行，因此新增片段提供了更多关于共享身体和接触动力学的证据，而无需增加新的容量。这与达到目标回报所需的计算量缓慢增长的情况一致（§4.1）。

想象加速了 rollout。在策略优化期间，推演速度决定了更新成本。图5(a) 显示，在实际使用的批量大小下，世界模型的运行速度比 MuJoCo Warp 快超过 10× 倍，而在 64k 并行环境下最高可达 62× 倍。

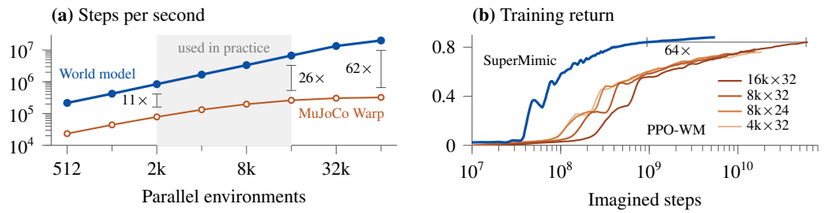


Figure 5: **Imagined rollouts are fast and differentiable.** (a) On an H100, over the batch sizes used in practice (shaded, 2k–16k parallel environments), the world model runs more than $10\times$ faster than MuJoCo Warp; the gap widens to $62\times$ at 64k, where MuJoCo Warp has flattened. One step is one 50 Hz control step, which MuJoCo Warp computes as four 5 ms substeps; the world model runs at the TF32 precision used in SuperMimic's training. (b) On walk/run/sprint motions from LAFAN1 [28], SuperMimic, trained with its actor–critic update, reaches the best training return of PPO trained on world-model rollouts (PPO-WM) with about $64\times$ fewer imagined steps. PPO-WM runs at four batch sizes (parallel rollouts $\times$ steps per rollout); 16k×32 is the best. Each curve is a centered moving average over ± 0.05 decade of imagined steps.

First-order gradients need fewer imagined steps. As a mechanism ablation, Figure 5(b) isolates the benefit of first-order policy updates: SuperMimic reaches the best PPO-WM return with about $64\times$ fewer imagined steps, so the gain is not just rollout speed.

4.3 A critic enables shorter horizons and push recovery

Figure 4(b, c) sweeps H with and without the critic. Without the critic, training fails at $H \leq 16$. With it, H=16 matches the critic-free H=32, and H=32 reaches a return of 0.82 fastest. Raising H from 64 to 128 nearly triples the time to that return and lowers the final return; H=128 is the longest horizon that fits on an H100. At matched return, the critic still changes behavior. Under 0.5–4 m/s velocity pushes while holding a static T-pose in simulation, the critic policy settles within 0.5–2.4 s. The critic-free policy never settles: it keeps stumbling even at 0.5 m/s and falls at 4 m/s.

4.4 Real-robot transfer

One general motion tracker deploys on a real Unitree G1 without per-motion tuning (Figure 6). The four motions cover locomotion (a skipping jump), contact-rich whole-body motion with the torso and limbs on the floor (a side roll), and acrobatics (a 90° flip and an in-place 360° flip).

4.5 Downstream tasks: RL encoder vs. distillation baseline

SuperMimic trains downstream goal encoders jointly with the motion-tracking interface, sharing one decoder and world model (Appendix B). Our study of downstream interfaces is preliminary. We train one velocity encoder on a velocity-tracking reward through the world model. We compare it with a distillation baseline: a velocity encoder trained on the motion-tracking reward, with commands read from the reference motions. Figure 7 evaluates both on 672 step responses spanning the velocity space (Appendix G). The RL encoder has the lower steady-state error on 650 of them, and the distillation encoder on 15. Only the distillation encoder falls, on seven fast turns with almost no motion-library data.

5 Conclusion

SuperMimic removes compute as the bottleneck to scaling humanoid whole-body control. A single learned, differentiable world model captures the dynamics every motion shares, and backpropagation-through-time through it amortizes a whole dataset into one policy. The policy trains on a single GPU, surpasses the return of a policy trained on a 128-GPU cluster, and transfers to a real Unitree G1. Its training cost grows slowly over a controlled dataset-length sweep. The

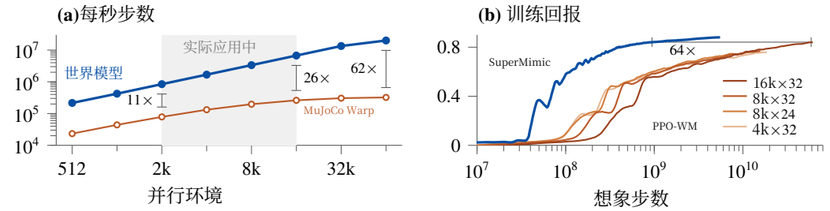


图 5：**想象 rollout 快速且可微**。(a) 在 H100 上，针对实际使用的批量大小（阴影区域，2k–16k 并行环境），世界模型的运行速度比 MuJoCo Warp 快 $10\times$ 倍以上；当达到 64k 时，差距扩大至 $62\times$ 倍，此时 MuJoCo Warp 已趋于平缓。一步指一个50Hz 控制步，MuJoCo Warp 将其计算为四个5 ms 子步；世界模型以 SuperMimic 训练中使用的 TF32 精度运行。(b) 在 LAFANSuperMimic 的行走/跑步/冲刺动作上，经过演员 - 评论家更新训练的 SuperMimic，仅需约 $64\times$ 更少的想象步数，即可达到基于世界模型 rollout 训练的 PPO（PPO-WM）的最佳训练回报。PPO-WM 在四种批量大小下运行（并行 rollout 数为 $\times$，每次 rollout 的步数为）；其中 16k表现最佳。每条曲线均为跨越 ± 0.05 个数量级想象步数的中心移动平均值。

一阶梯度需要更少的想象步数。作为机制消融实验，图5(b) 隔离了一阶策略更新的优势：SuperMimic 以大约64×更少的想象步数达到了最佳的 PPO-WM 回报，因此该增益不仅仅来自推演速度。

4.3 critic 支持更短的 horizon 并实现推扰恢复

图 4(b, c) 展示了有无 critic 时 H 的扫描结果。若无 critic，训练在 $H \leq 16$ 处失败；若有 critic，H=16 能与无 critic 的 H=32 相匹配，且 H=32 最快达到 0.82 的回报。将 H 从 64 提升至 128 几乎使达到该回报所需的时间增加两倍，并降低了最终回报；H=128 是 H100 上能容纳的最长 horizon。在回报相当的情况下，critic 仍会改变行为表现。在模拟中，当机器人保持静态 T 姿势并承受 0.5–4 m/s 的速度推扰时，基于 critic 的策略可在 0.5–2.4 秒内稳定下来。而无 critic 的策略始终无法稳定：即使在 0.5 m/s 的推扰下也会不断踉跄，并在 4 m/s 时摔倒。

4.4 真实机器人迁移

一个通用运动追踪器可直接部署于真实的 Unitree G1 机器人，无需针对每个动作进行调优（见图6）。这四种动作涵盖：运动（跳绳跳跃）、涉及躯干与四肢接触地面的丰富接触全身运动（侧滚），以及杂技动作（一次 90° 翻转和一次原地 360° 翻转）。

4.5 下游任务：强化学习编码器与蒸馏基线的对比

SuperMimic 将下游目标编码器与运动追踪接口联合训练，共享一个解码器和世界模型（附录 B）。我们对下游接口的研究尚属初步阶段。我们通过世界模型，基于速度跟踪奖励训练了一个速度编码器。我们将它与一个蒸馏基线进行比较：该基线是一个基于运动跟踪奖励训练的速度编码器，其指令从参考运动中读取。图7在涵盖整个速度空间的 672 个阶跃响应上对两者进行了评估（附录G）。其中，强化学习编码器在 650 个响应中具有更低的稳态误差，而蒸馏编码器仅在 15 个响应中表现更优。只有蒸馏编码器发生了摔倒，出现在七个快速转弯场景中，这些场景几乎没有任何运动库数据。

5 结论

SuperMimic 消除了计算资源作为扩展人形机器人全身控制的瓶颈。一个单一的可微世界模型捕捉了所有运动共享的动力学，并通过时间反向传播将整个数据集摊销到一个策略中。该策略在单个 GPU 上训练，其回报超过了在 128 GPU 集群上训练的策略，并成功迁移到真实的 Unitree G1 上。其训练成本在受控的数据集长度扫描中缓慢增长。



Figure 6: **One general motion tracker, trained through the learned world model, transfers to a real Unitree G1.** (a) Skipping jump, (b) side roll, (c) 90° flip, and (d) in-place 360° flip.

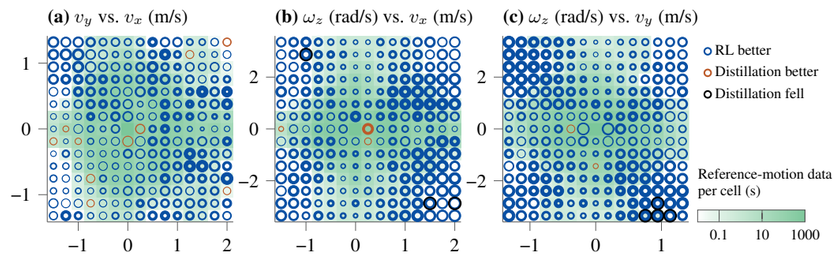


Figure 7: **The policy with the RL-trained velocity encoder tracks step commands more accurately than the distillation baseline.** Each panel is a plane of the command space with the third component at zero, and each ring is one command. The outer disc's area is proportional to the larger of the two policies' relative steady-state errors in the step response to that command, and the hole's area to the smaller one; the color marks the better policy. Details in Appendix G.

same model can train a downstream velocity interface directly on the interface's own task reward. Measuring the scaling laws of motion tracking is now affordable, at one GPU per run.

6 Limitations and Future Work

- **Multi-interface training is understudied.** We train the velocity encoder concurrently with the motion tracker, but the velocity-tracking reward updates only the velocity encoder and not the shared decoder (Appendix B). Concurrent training is therefore the same as training the velocity encoder after the motion tracker, and might not be necessary. Whether concurrent training helps once the velocity-tracking reward also updates the decoder is untested.
- **Simulator-only dynamics.** The world model learns only from simulator transitions. Real-robot data could refine it, and its differentiability could support system identification.
- **No terrain or object interaction.** SuperMimic trains only on flat ground. Extending it to terrain and object interaction is ongoing work.



图 6：**一个通用运动追踪器，通过所学的世界模型训练，成功迁移至真实的 Unitree G1**。(a) 跳绳跳跃，(b) 侧滚，(c) 90° 翻转，以及 (d) 原地 360° 翻转。

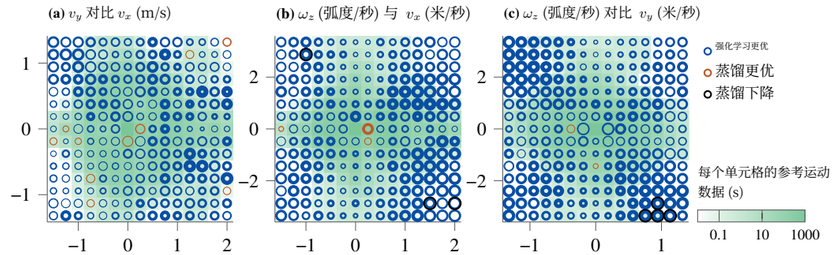


图 7：**采用强化学习训练的速度编码器的策略，比蒸馏基线更准确地跟踪阶跃指令**。每个面板是指令空间的一个平面，其中第三个分量为零，每个圆环代表一个指令。外圆盘的面积与两个策略在该指令阶跃响应中的较大相对稳态误差成正比，内孔的面积与较小的误差成正比；颜色标记表现更优的策略。详见附录G。

同一模型可以直接基于接口自身的任务奖励，训练下游的速度接口。现在，以每次运行仅需一个 GPU 的成本，即可负担得起地测量运动跟踪的缩放定律。

6 局限性与未来工作

- **多接口训练研究不足**。我们并发训练速度编码器与运动追踪器，但速度跟踪奖励仅更新速度编码器，而不更新共享解码器（附录B）。因此，并发训练等同于在运动追踪器之后训练速度编码器，可能并非必要。一旦速度跟踪奖励也更新解码器，并发训练是否有益尚未测试。
- **仅限模拟器的动力学**。世界模型仅从仿真器转移中学习。真实机器人数据可对其加以优化，而其可微性可支持系统辨识。
- **无地形或物体交互**。SuperMimic 仅在平地上进行训练。将其扩展至地形和物体交互正在进行中。

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A Instantiation: observation, action, reward, and sim context

This appendix instantiates the shared spaces of the POMDP of §3.1. The motion-tracking interface follows the BeyondMimic formulation [3]; the changes we make are noted inline.

Observation. The proprioceptive observation $\boldsymbol{o}_t$ comprises the joint positions relative to the default pose $\boldsymbol{q} - \boldsymbol{q}^0$, the joint velocities $\dot{\boldsymbol{q}}$, the base angular velocity from the inertial measurement unit (IMU), the projected gravity in the base frame, and the previous action $\boldsymbol{a}_{t-1}$ (Table A.1); the policy receives the last 10 observations.

Observation Term	Dim.	Noise η_t	Unit
Joint positions $\boldsymbol{q} - \boldsymbol{q}^0$	29	$\mathcal{U}[-0.01, 0.01]$	rad
Joint velocities $\dot{\boldsymbol{q}}$	29	$\mathcal{U}[-0.5, 0.5]$	rad/s
Base angular velocity (IMU)	3	$\mathcal{U}[-0.2, 0.2]$	rad/s
Projected gravity	3	$\mathcal{U}[-0.05, 0.05]$	–
Previous action $\boldsymbol{a}_{t-1}$	29	–	–

Table A.1: **Proprioceptive observation $\boldsymbol{o}_t$.** The noise column gives the per-dimension sensor noise $\boldsymbol{\eta}_t \sim \rho_\eta$.

Goal. For the motion-tracking interface, the goal $\boldsymbol{g}_t$ is read from the reference motion. Compared with BeyondMimic [3], we condition the encoder on a $K = 10$ future reference sequence rather than a single-frame target. Other goal interfaces replace $\boldsymbol{g}_t$ with their corresponding command representation, such as target velocity commands (Appendix B).

Action. The action $\boldsymbol{a}_t$ maps to a joint setpoint $\boldsymbol{q}^{\text{sp}} = \boldsymbol{\alpha}\, \boldsymbol{a}_t + \boldsymbol{q}^0$ about the default pose $\boldsymbol{q}^0$. Following BeyondMimic, the low PD gains make this setpoint a proxy for joint torque rather than a desired position.

Reward. We use BeyondMimic's motion-tracking reward (Table A.2), removing its self-collision penalty. That penalty guards against reward hacking by a stochastic policy. When the reference brings two bodies close without contact, pressing them together suppresses the action noise and raises the velocity-tracking reward. Our policy is deterministic in imagined rollouts (§3.3), so the exploit has no payoff and the term is unnecessary. The joint-limit penalty is lowered from -10 to -2.

Termination. An episode terminates when the root body deviates from the reference beyond a height or tilt threshold ($|\boldsymbol{e}_{p,z}| > 0.25\,\text{m}$ or $e_{\text{tilt}} > 0.8$). The tilt error e_{tilt} is the absolute difference between the z-components of the gravity direction expressed in the reference and robot root (pelvis) frames. For an upright reference, e_{tilt} reaches 0.8 at about 78° of tilt. Unlike BeyondMimic, we do not terminate on end-effector deviation.

Sim context and domain randomization. The per-episode sim context $\boldsymbol{\xi}$ collects the domain-randomization parameters fed to the world model (§3.2); Table A.3 lists them. We omit BeyondMimic's root-velocity perturbations: such pushes inject state discontinuities that the deterministic world model cannot capture, and we find randomization over the physical parameters alone suffices for transfer to hardware.

B Joint training across goal interfaces

In §3, the goal $\boldsymbol{g}_t$ is read from a full reference motion. Downstream goal interfaces specify the motion only in part or in another form. The same dynamics model can train several goal interfaces because the body physics is shared. We instantiate one POMDP $\mathcal{M}_i$ per goal interface (for example, a full-reference motion target, a velocity command, end-effector poses, or an SMPL body-pose target), each with its own goal space $\mathcal{G}_i$, reward r_i, and termination d_i, and train each with its own

A 实例化：观测、动作、奖励和仿真上下文

本附录实例化了§3.1中部分可观察马尔可夫决策过程（POMDP）的共享空间。运动追踪接口遵循 BeyondMimic 公式 [3]；我们所做的更改已在文中标注。

观测。本体感觉观测 $\boldsymbol{o}_t$ 包含相对于默认姿态的关节位置 $\boldsymbol{q} - \boldsymbol{q}^0$、关节速度 $\dot{\boldsymbol{q}}$、来自惯性测量单元（IMU）的基座角速度、基座坐标系中的投影重力，以及上一动作 $\boldsymbol{a}_{t-1}$（表A.1）；策略接收最近 10 次观测。

观测项	维度	噪声 η_t	Unit
关节位置 $\boldsymbol{q} - \boldsymbol{q}^0$	29	$\mathcal{U}[-0.01, 0.01]$	rad
关节速度 $\dot{\boldsymbol{q}}$	29	$\mathcal{U}[-0.5, 0.5]$	rad/s
基座角速度（IMU）	3	$\mathcal{U}[-0.2, 0.2]$	rad/s
投影重力	3	$\mathcal{U}[-0.05, 0.05]$	–
上一动作 $\boldsymbol{a}_{t-1}$	29	–	–

表 A.1：**本体感觉观测 $\boldsymbol{o}_t$**。噪声列给出了每维度的传感器噪声$\boldsymbol{\eta}_t \sim \rho_\eta$。

目标。对于运动追踪接口，目标 $\boldsymbol{g}_t$ 是从参考动作中读取的。与 BeyondMimic [3], 相比，我们将编码器条件化为 $K = 10$ 未来参考序列，而非单帧目标。其他目标接口将 $\boldsymbol{g}_t$ 替换为其对应的指令表示，例如目标速度指令（附录B）。

动作。动作 $\boldsymbol{a}_t$ 映射为关于默认姿态 $\boldsymbol{q}^0$ 的关节设定点 $\boldsymbol{q}^{\text{sp}} = \boldsymbol{\alpha}\, \boldsymbol{a}_t + \boldsymbol{q}^0$。遵循 BeyondMimic 的做法，较低的 PD 增益使得该设定点成为关节力矩的代理，而非期望位置。

奖励。我们使用 BeyondMimic 的运动跟踪奖励（表A.2），并移除了其自碰撞惩罚。该惩罚旨在防止随机策略进行奖励作弊。当参考动作使两个物体靠近但未接触时，将它们压在一起会抑制动作噪声并提高速度跟踪奖励。我们的策略在想象 rollout 中是确定性的（§3.3），因此该 exploit 无法带来收益，该项变得不必要。关节限位惩罚从 -10降低至 -2。

终止条件。当根部位偏离参考值超过高度或倾斜阈值（$|\boldsymbol{e}_{p,\ z}| > 0.25$米或 $e_{\text{tilt}} > 0.8$）时，回合终止。倾斜误差 e_{tilt} 是重力方向在参考坐标系与机器人根部（骨盆）坐标系中表示的 z 分量之间的绝对差值。对于直立参考姿态，当倾斜角度约为 78° 时，e_{tilt} 达到 0.8。与 BeyondMimic 不同，我们不会因末端执行器偏差而终止回合。

仿真上下文和域随机化。每回合的仿真上下文 $\boldsymbol{\xi}$ 收集提供给世界模型（§3.2）的域随机化参数；表A.3列出了这些参数。我们省略了 BeyondMimic 中的根速度扰动：此类推力会注入状态不连续性，而确定性世界模型无法捕捉这些不连续性；我们发现仅对物理参数进行随机化就足以实现向硬件的迁移。

B 跨目标接口的联合训练

在§3中，目标 $\boldsymbol{g}_t$ 是从完整的参考动作中读取的。下游目标接口仅部分地或以另一种形式指定动作。相同的动力学模型可以训练多个目标接口，因为身体物理是共享的。我们为每个目标接口实例化一个部分可观察马尔可夫决策过程 $\mathcal{M}_i$（例如，完整参考动作目标、速度指令、末端执行器姿态或 SMPL 人体姿态目标），每个接口都有其各自的目标空间 $\mathcal{G}_i$、奖励 r_i和终止条件 d_i，并分别使用其各自的

Reward Term	Equation	Weight
Task (tracking)		
Body Position	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\mathbf{p}_b^{\text{des}}-\mathbf{p}_b\rVert^2\big)/0.3^2\Big)$	1.0
Body Orientation	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\log(R_b^{\text{des}}R_b^\top)\rVert^2\big)/0.4^2\Big)$	1.0
Body Linear velocity	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\mathbf{v}_b^{\text{des}}-\mathbf{v}_b\rVert^2\big)/1.0^2\Big)$	1.0
Body Angular velocity	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\boldsymbol{\omega}_b^{\text{des}}-\boldsymbol{\omega}_b\rVert^2\big)/3.14^2\Big)$	1.0
Root Position	$\exp\Big(-\lVert\mathbf{p}_{\text{root}}^{\text{des}}-\mathbf{p}_{\text{root}}\rVert^2/0.3^2\Big)$	0.5
Root Orientation	$\exp\Big(-\lVert\log(R_{\text{root}}^{\text{des}}R_{\text{root}}^\top)\rVert^2/0.4^2\Big)$	0.5
Regularization		
Action smoothness	$\lVert\mathbf{a}_t-\mathbf{a}_{t-1}\rVert^2$	-0.1
Joint limit	$\sum_j\big[\max(0,\,q_j-q_j^{\max})+\max(0,\,q_j^{\min}-q_j)\big]$	-2

Table A.2: **Motion-tracking interface reward, following BeyondMimic [3] with the self-collision penalty removed and the joint-limit weight lowered from -10 to -2 (§3.3).** $\mathcal{T}$ is the set of tracked bodies.

Domain Randomization	Sampling Distribution
Friction coefficients	$\mu\sim\mathcal{U}[0.3,\,1.2]$
Joint calibration offsets [rad]	$\Delta q_j^0\sim\mathcal{U}[-0.01,\,0.01]$
Torso's center-of-mass offset [m]	$\Delta x\sim\mathcal{U}[-0.025,0.025]$, $\Delta y\sim\mathcal{U}[-0.05,0.05]$, $\Delta z\sim\mathcal{U}[-0.05,0.05]$

Table A.3: **Domain-randomization parameters that make up the sim context ξ, following BeyondMimic [3] with the root-velocity perturbations removed.** ($\mathcal{U}[\cdot]$: uniform distribution.)

critic V_{ψ_i}. A single policy network uses per-interface goal encoders $\text{enc}_{\theta,i}$ and a shared decoder dec_θ,

$$\boldsymbol{z}=\text{enc}_{\theta,i}(\boldsymbol{g})\in\mathcal{Z},\qquad \boldsymbol{a}=\text{dec}_\theta(\boldsymbol{z},\boldsymbol{o}),\qquad \boldsymbol{g}\in\mathcal{G}_i, \tag{A.1}$$

and training optimizes the lifted objective $\max_\theta\mathbb{E}_{i\sim p(\mathcal{I})}[\hat{J}_i(\theta)]$, where $p(\mathcal{I})$ is a distribution over the set $\mathcal{I}$ of goal interfaces. Each interface i thus has its own policy $\boldsymbol{a}=\pi_i(\boldsymbol{o},\boldsymbol{g}):=\text{dec}_\theta(\text{enc}_{\theta,i}(\boldsymbol{g}),\boldsymbol{o})$; the velocity encoder also receives a feature of the observation history (Table A.4). Each $\hat{J}_i$ is the objective of Eq. 2 with π_i in place of π_θ, the reward and termination of $\mathcal{M}_i$, and the critic V_{ψ_i}. Every interface trains through the same dynamics model f_ω, which is not retrained per interface.

A downstream interface is trained by distillation or by RL, depending on its reward. Distillation uses the motion-tracking reward (§3.3), with the command read from the reference motion. The reward gradient then trains that interface to reproduce the reference motion from this command. RL optimizes the interface's own task reward, such as velocity tracking, directly through the world model. On a downstream interface's rollouts, the return gradient passes through the shared decoder to that interface's encoder, with a stop-gradient on the decoder's weights; the decoder is still trained by the motion-tracking objective. The RL-trained velocity encoder of §4.5 also has a latent-alignment loss toward the motion encoder's latent on rollouts with reference commands (Appendix G).

C Network architectures

Table A.4 lists the world-model dynamics network of §3.2 and the policy and critic networks of Appendix B.

奖励项	方程	权重
任务（跟踪）		
身体位置	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\mathbf{p}_b^{\text{des}}-\mathbf{p}_b\rVert^2\big)/0.3^2\Big)$	1.0
身体朝向	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\log(R_b^{\text{des}}R_b^\top)\rVert^2\big)/0.4^2\Big)$	1.0
身体线速度	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\mathbf{v}_b^{\text{des}}-\mathbf{v}_b\rVert^2\big)/1.0^2\Big)$	1.0
身体角速度	$\exp\Big(-\big(\frac{1}{\lvert\mathcal{T}\rvert}\sum_{b\in\mathcal{T}}\lVert\boldsymbol{\omega}_b^{\text{des}}-\boldsymbol{\omega}_b\rVert^2\big)/3.14^2\Big)$	1.0
根部位姿	$\exp\Big(-\lVert\mathbf{p}_{\text{root}}^{\text{des}}-\mathbf{p}_{\text{root}}\rVert^2/0.3^2\Big)$	0.5
根部朝向	$\exp\Big(-\lVert\log(R_{\text{root}}^{\text{des}}R_{\text{root}}^\top)\rVert^2/0.4^2\Big)$	0.5
正则化		
动作平滑性	$\lVert\mathbf{a}_t-\mathbf{a}_{t-1}\rVert^2$	-0.1
关节限制	$\sum_j\big[\max(0,\quad q_j-q_j^{\max})+\max(0,\quad q_j^{\min}-q_j)\big]$	-2

表 A.2：**运动追踪接口奖励，遵循 BeyondMimic [3]，移除了自碰撞惩罚并将关节限制权重从 -10 降低至 -2**（§3.3）。 $\mathcal{T}$为被追踪刚体集合。

域随机化	采样分布
摩擦系数	$\mu\sim\mathcal{U}[0.3,\,1.2]$
关节校准偏移 [rad]	$\Delta q_j^0\sim\mathcal{U}[-0.01,\,0.01]$
躯干质心偏移 [m]	$\Delta x\sim\mathcal{U}[-0.025,0.025]$, $\Delta y\sim\mathcal{U}[-0.05,0.05]$, $\Delta z\sim\mathcal{U}[-0.05,0.05]$

表 A.3：**构成仿真上下文 ξ的域随机化参数，遵循 BeyondMimic [3] 但移除了根速度扰动。**（$\mathcal{U}[\cdot]$: 均匀分布。）

critic V_{ψ_i}。单个策略网络使用每接口的目标编码器 $\text{enc}_{\theta,\,i}$ 以及一个共享解码器 dec_θ,

$$\boldsymbol{z}=\text{enc}_{\theta,i}(\boldsymbol{g})\in\mathcal{Z},\qquad \boldsymbol{a}=\text{dec}_\theta(\boldsymbol{z},\boldsymbol{o}),\qquad \boldsymbol{g}\in\mathcal{G}_i, \tag{A.1}$$

且训练优化提升目标$\max_\theta\mathbb{E}_{i\sim p(\mathcal{I})}[\hat{J}_i(\theta)]$，其中 $p(\mathcal{I})$ 是目标接口集合 $\mathcal{I}$ 上的分布。因此，每个接口 i 都有其自身的策略 $\boldsymbol{a}=\pi_i(\boldsymbol{o},\boldsymbol{g}):=\text{dec}_\theta(\text{enc}_{\theta,i}(\boldsymbol{g}),\boldsymbol{o})$;˄速度编码器还接收观测历史的一个特征（表 A.4）。每个 J_i 是公式2 中的目标，其中用 π_i 替代 π_θ，并采用 $\mathcal{M}_i$的奖励和终止条件，以及 critic V_{ψ_i}。所有接口均通过同一个动力学模型 f_ω进行训练，该模型不会针对每个接口重新训练。

下游接口通过蒸馏或强化学习进行训练，具体取决于其奖励。蒸馏利用运动跟踪奖励（§3.3），其中指令从参考动作中读取。随后，奖励梯度训练该接口以根据此指令复现参考动作。强化学习则直接通过世界模型优化接口自身的任务奖励，例如速度跟踪。在下游接口的 rollout 过程中，回报梯度穿过共享解码器传递至该接口的编码器，同时对解码器的权重施加梯度停止；解码器仍由运动跟踪目标进行训练。§4.5 中所述的强化学习训练的速度编码器，在使用参考指令的 rollout 中，还具有针对运动编码器隐变量的隐变量对齐损失（附录 G）。

C 网络架构

表A.4 列出了§3.2 中的世界模型动力学网络以及附录B中的策略和 critic 网络。

Network	Input	Output	Architecture
Dynamics f_ω	$\hat{\boldsymbol{s}}_t,\ \boldsymbol{a}_t,\ \boldsymbol{\xi}$	$\hat{\boldsymbol{s}}_{t+1}$	MLP (2048, 2048, 2048)
Motion encoder $\mathrm{enc}_{\theta,\mathrm{motion}}$	K=10 future reference frames	$\boldsymbol{z}$	MLP (2048, 1024, 512, 256)
Velocity encoder $\mathrm{enc}_{\theta,\mathrm{vel}}$ (Appendix G)	normalized command (v_x, v_y, ω_z), feature of the last 10 frames of $\boldsymbol{o}$	$\boldsymbol{z}$	MLP (256, 256)
Shared decoder dec_θ	$\boldsymbol{z}$, last 10 frames of $\boldsymbol{o}$	$\boldsymbol{a}_t$	MLP (2048, 1024, 512, 256)
Critic V_{ψ_i}	$\hat{\boldsymbol{s}}_t,\ \boldsymbol{g}_t,\ \boldsymbol{\xi}$	scalar	MLP (256, 128); one critic per interface

Table A.4: **World-model, policy, and critic networks.** MLP: multilayer perceptron, listed by hidden widths. The latent $\boldsymbol{z}$ is 32-dimensional and bounded by $\tanh$.

Algorithm A.1 SuperMimic training. Both ω and θ update every iteration; ω is held fixed (stop-gradient) only inside the policy's backward pass. For a downstream interface, $\hat{J}_i$ updates only its encoder $\mathrm{enc}_{\theta,i}$.

Require: world model f_ω; policy π_θ shared across interfaces $\{\mathcal{M}_i\}_{i\in\mathcal{I}}$; replay buffer $\mathcal{B}$
for each iteration **do**
  ▷ **Collect.** Roll out π_θ with action noise in simulation; push transitions to $\mathcal{B}$, replacing the oldest
    Draw reset start points by return-based sampling ▷ Appendix D.1
  ▷ **Fit world model.** Sample a logged k-step window from $\mathcal{B}$, roll f_ω out autoregressively on the logged actions, and step $\omega \leftarrow \omega - \alpha_\omega \nabla_\omega \mathcal{L}_{\mathrm{WM}}$ ▷ Eq. 1
  ▷ **Improve policy.** Sample an interface $i \sim p(\mathcal{I})$ and a start state from $\mathcal{B}$
    Imagine H steps on-policy through the fixed f_ω: $\hat{\boldsymbol{s}}_{t+1} = f_\omega\big(\hat{\boldsymbol{s}}_t,\ \pi_i(\tilde{\boldsymbol{o}}_t + \boldsymbol{\eta}_t, \boldsymbol{g}_t),\ \boldsymbol{\xi}\big)$
    Step $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \hat{J}_i$ ▷ Eq. 2; analytic gradients through f_ω; critic V_{ψ_i}
    Fit the critic V_{ψ_i} to the GAE targets of the imagined rollout ▷ Eq. 3
end for

D Training loop and hyperparameters

D.1 Algorithm

Algorithm A.1 summarizes the SuperMimic training loop of §3.1; its three steps correspond one-to-one to the panels of Figure 2.

Return-based reset sampling. Simulator episodes reset to points of the motion library, favoring those where the policy currently earns low return. The motion timeline is divided into 1 s bins, and each bin b keeps a running estimate G_b of the discounted return from it, updated on the policy's own rollouts. Resets mix three distributions: 50% uniform over bins, 40% from $\mathrm{softmax}_b(-25\, G_b/G_{\max})$, where $G_{\max}$ is the largest attainable discounted return, and 10% in proportion to each bin's past softmax weight, so a hard motion keeps some practice after it is learned.

D.2 Hyperparameters

Table A.5 lists the hyperparameters. Where they differ, values are given for the one-GPU and the two-GPU run of Table A.8.

E Motion data filtering

We train on the publicly available G1-retargeted BONES-SEED motion set [27]. The raw release contains 142,220 clips, but many interact with terrain or objects the robot cannot use (stairs, boxes, ladders, chairs) or carry retargeting artifacts that make them infeasible for a free-standing humanoid on flat ground. We remove these in two passes and keep 129,514 clips (Table A.7).

Name filter. Each clip is named by its action. We drop any clip whose name contains a term denoting contact with terrain or an object the robot cannot use—e.g. `stair`, `ladder`, `box`, `chair`,

网络	输入	输出	架构
动力学 f_ω	$\hat{\boldsymbol{s}}_t,\ \boldsymbol{a}_t,\ \boldsymbol{\xi}$	$\hat{\boldsymbol{s}}_{t+1}$	多层感知机 (2048, 2048, 2048)
运动编码器 $\mathrm{enc}_{\theta,\mathrm{motion}}$	K=10未来参考帧	$\boldsymbol{z}$	多层感知机 (2048, 1024, 512, 256)
速度编码器 $\mathrm{enc}_{\theta,\mathrm{vel}}$（附录 G）	归一化指令 $(v_x,\ v_y,\ \omega_z)$，最后 10 帧的 v_x 特征	$\boldsymbol{z}$	多层感知机 (256, 256)
共享解码器 dec_θ	$\boldsymbol{z}$，$\boldsymbol{o}$的最后 10 帧	$\boldsymbol{a}_t$	多层感知机 (2048, 1024, 512, 256)
Critic V_{ψ_i}	$\hat{\boldsymbol{s}}_t,\ \boldsymbol{g}_t,\ \boldsymbol{\xi}$	标量	多层感知机 (256, 128)；每个对应一个 critic 接口

表 A.4：**世界模型、策略和 critic 网络**。MLP：多层感知机，按隐藏层宽度列出。隐变量 $\boldsymbol{z}$ 为 32 维，并由双曲正切界定。

Algorithm A.1 SuperMimic training. Both ω and θ update every iteration; ω is held fixed (stop-gradient) only inside the policy's backward pass. For a downstream interface, $\hat{J}_i$ updates only its encoder $\mathrm{enc}_{\theta,i}$.

Require: world model f_ω; policy π_θ shared across interfaces $\{\mathcal{M}_i\}_{i\in\mathcal{I}}$; replay buffer $\mathcal{B}$
for each iteration **do**
  ▷ **Collect.** Roll out π_θ with action noise in simulation; push transitions to $\mathcal{B}$, replacing the oldest
    Draw reset start points by return-based sampling ▷ Appendix D.1
  ▷ **Fit world model.** Sample a logged k-step window from $\mathcal{B}$, roll f_ω out autoregressively on the logged actions, and step $\omega \leftarrow \omega - \alpha_\omega \nabla_\omega \mathcal{L}_{\mathrm{WM}}$ ▷ Eq. 1
  ▷ **Improve policy.** Sample an interface $i \sim p(\mathcal{I})$ and a start state from $\mathcal{B}$
    Imagine H steps on-policy through the fixed f_ω: $\hat{\boldsymbol{s}}_{t+1} = f_\omega\big(\hat{\boldsymbol{s}}_t,\ \pi_i(\tilde{\boldsymbol{o}}_t + \boldsymbol{\eta}_t, \boldsymbol{g}_t),\ \boldsymbol{\xi}\big)$
    Step $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \hat{J}_i$ ▷ Eq. 2; analytic gradients through f_ω; critic V_{ψ_i}
    Fit the critic V_{ψ_i} to the GAE targets of the imagined rollout ▷ Eq. 3
end for

D 训练循环与超参数

D.1 算法

算法A.1总结了 § 3.1中的 SuperMimic 训练循环；其三个步骤与图2中的各个面板一一对应。

基于回报的复位采样。模拟器回合复位至动作库中的点，优先选择策略当前获得低回报的位置。动作时间线被划分为 1 秒的区间，每个区间 b 维护一个对其折扣回报的运行估计 G_b，该估计根据策略自身的 rollout 进行更新。复位混合了三种分布：50% 在区间上均匀分布，40% 来自 $\mathrm{softmax}_b(-25\, G_b/G_{\max})$（其中 $G_{\max}$ 为可达到的最大折扣回报），10% 与各区间过去的 softmax 权重成比例，从而确保高难度动作在学习后仍能保留一定的练习机会。

D.2 超参数

表A.5 列出了超参数。在存在差异的情况下，分别给出了表A.8中单 GPU 和双 GPU 运行的数值。

E 运动数据过滤

我们在公开可用的 G1 重定向 BONES-SEED 动作集上进行训练 [27]。原始发布版本包含142,220 个片段，但其中许多涉及机器人无法使用的地形或物体（如楼梯、箱子、梯子、椅子），或包含重定向伪影，使其不适用于在平地上自由站立的人形机器人。我们分两步移除这些片段，最终保留129,514 个片段（表A.7）。

名称过滤器。每个片段以其动作命名。我们会丢弃任何名称中包含表示与地形或机器人无法使用的物体接触术语的片段，例如楼梯、梯子、箱子、椅子，

Hyperparameter	Symbol	Value
World model		
Window length	k	8
Optimizer		Adam
Learning rate	α_ω	1 GPU: 2×10^{-4}; 2 GPUs: $\sqrt{2} \times 2 \times 10^{-4}$
Batch size		10 batches of 2048 windows per update
Gradient clipping		global norm 1.0
Imagination and policy		
Imagination horizon	H	32
Optimizer		Adam, $\beta = (0.9,\ 0.95)$
Learning rate	α_θ	same as α_ω
Batch size		4096 imagined rollouts per update
Gradient clipping		global norm 1.0, then element-wise 0.1
Discount	γ	0.99
Sensor noise	$\boldsymbol{\eta}_t \sim \rho_\eta$	Table A.1
Critic		
GAE parameter	λ	0.95
Optimizer		Adam
Learning rate		same as α_ω
Gradient clipping		global norm 1.0
Critic epochs per update		5, each over 4 shuffled minibatches
Data collection		
Simulator environments per GPU		4096
Action noise	$\boldsymbol{\epsilon}_t \sim \rho_\epsilon$	$\mathcal{N}(0,\ 0.1^2)$ per action dimension
Replay-buffer size		1000 steps per environment
Schedule		
Update-to-data ratio		0.4
Policy updates per world-model update		1
Learning-rate schedule		cosine decay to $0.03\times$ over 250,000 iterations, then constant
Training length		320,000 iterations
Multi-GPU		data parallel; gradients averaged across GPUs

Table A.5: **SuperMimic hyperparameters.**

Reward Term	Equation	Weight
Task (velocity tracking)		
Linear velocity	$\exp\Big(-\lVert\mathbf{v}_{xy} - \mathbf{v}_{xy}^{\mathrm{cmd}}\rVert^2 / 0.5^2\Big)$	2.0
Yaw rate	$\exp\Big(-(\omega_z - \omega_z^{\mathrm{cmd}})^2 / 0.5\Big)$	2.0
Regularization		
Action smoothness	$\lVert\mathbf{a}_t - \mathbf{a}_{t-1}\rVert^2$	-0.3
Joint limit	$\sum_j \left[\max(0,\ q_j - q_j^{\max}) + \max(0,\ q_j^{\min} - q_j)\right]$	-2
Torso sideways lean	$\sin^2 \phi_{\mathrm{torso}}$	-4
Torso roll and pitch rate	$\omega_{\mathrm{roll}}^2 + \omega_{\mathrm{pitch}}^2$	-0.05
Termination (fall)		
$h_{\mathrm{root}} < 0.35\,\mathrm{m}$ or $1 + u_z > 0.7$		

Table A.6: **Velocity-tracking reward and termination used to train the RL encoder.** $\mathbf{v}_{xy}$ is the planar root velocity in the heading frame, ω_z the world yaw rate, and $(\mathbf{v}_{xy}^{\mathrm{cmd}}, \omega_z^{\mathrm{cmd}})$ the command. ϕ_{torso} is the sideways lean of the torso z-axis (forward lean is not penalized) and $\omega_{\mathrm{roll}}, \omega_{\mathrm{pitch}}$ the torso roll and pitch rates, all in the heading frame. h_{root} is the root height and u_z the z-component of the unit gravity direction in the root frame, so $1 + u_z$ is 0 upright and 2 upside down, and the tilt condition triggers at about 73°. Every term, here and in Table A.2, is multiplied by the control period $\Delta t = 0.02\,\mathrm{s}$.

超参数	符号	值
世界模型		
窗口长度	k	8
优化器		Adam
学习率	α_ω	1 GPU： 2×10^{-4}；2 GPUs： $\sqrt{2} \times 2 \times 10^{-4}$
批量大小		每次更新包含 10 个批次，每批次 2048 个窗口
梯度裁剪		全局范数 1.0
想象与策略		
想象范围	H	32
优化器		Adam, $\beta = (0.9,\ 0.95)$
学习率	α_θ	与 α_ω 相同
批量大小		每次更新 4096 次想象 rollout
梯度裁剪		全局范数 1.0，然后逐元素 0.1
折扣	γ	0.99
传感器噪声	$\boldsymbol{\eta}_t \sim \rho_\eta$	表A.1
critic		
GAE 参数	λ	0.95
优化器		Adam
学习率		与 α_ω相同
梯度裁剪		全局范数 1.0
每次更新 Critic 轮数		5，每轮在 4 个打乱的小批量上进行
数据收集		
每 GPU 模拟器环境数		4096
动作噪声	$\boldsymbol{\epsilon}_t \sim \rho_\epsilon$	$\mathcal{N}(0,\ 0.1_2)$ 每动作维度
回放缓冲区大小		每环境 1000 步数
调度		
更新数据比		0.4
每次世界模型更新的策略更新次数		1
学习率调度		余弦衰减至 $0.03\times$，超过 250,000 迭代次数，然后保持恒定
训练长度		320,000 次迭代
多 GPU		数据并行；梯度在多个 GPU 之间平均

表 A.5：**SuperMimic 超参数**。

奖励项	方程	权重
任务（速度跟踪）		
线速度	$\exp\Big(-\lVert\mathbf{v}_{xy} - \mathbf{v}_{xy}^{\mathrm{cmd}}\rVert^2 / 0.5^2\Big)$	2.0
偏航速率	$\exp\Big(-(\omega_z - \omega_z^{\mathrm{cmd}})^2 / 0.5\Big)$	2.0
正则化		
动作平滑性	$\lVert\mathbf{a}_t - \mathbf{a}_{t-1}\rVert^2$	-0.3
关节限制	$\sum_j \left[\max(0,\ q_j - q_j^{\max}) + \max(0,\ q_j^{\min} - q_j)\right]$	-2
躯干侧倾	$\sin^2 \phi_{\mathrm{torso}}$	-4
躯干横滚和俯仰速率	$\omega_{\text{横滚}}^2 + \omega_{\text{俯仰}}^2$	-0.05
终止条件（摔倒）		
$h_{\mathrm{root}} < 0.35$米或 $1 + u_z > 0.7$		

表 A.6：**用于训练强化学习编码器的速度跟踪奖励和终止条件**。$\mathbf{v}_{xy}$ 是航向坐标系中的平面根速度， ω_z 世界偏航速率和指令。ϕ_{torso} 是躯干的侧倾 z轴（前倾不会受到惩罚）， ω_{roll}, ω_{pitch} 躯干横滚和俯仰速率，均在航向坐标系中。 h_{root} 是根高度， u_z 单位重力方向在根坐标系中的 z分量，因此 $1 + u_z$ 在直立时为 0，倒立时为 2，倾斜条件在约 73°时触发。此处的每一项以及表A.2中的每一项，均乘以控制周期。 $\Delta t = 0.02\,\mathrm{s}$

Stage	Clips
Raw BONES-SEED (G1 retarget)	142,220
After name filter	129,667
After feasibility filter	129,514

Table A.7: **Number of BONES-SEED clips before filtering and after each filter.**

`sit`, `climb`, `jump_off_50cm`, `look_over_fence`. This pass (adapted from the keyword list in SONIC's released data-processing code [1], plus a few terms it misses) removes 12,553 clips.

Feasibility filter. The name filter misses clips whose names do not reveal the problem (e.g. a routine that walks onto a box). We run each motion through MuJoCo forward kinematics and flag it from the resulting body-link and foot-geometry trajectories, writing $z_{\min}(t) = \min_b z_b(t)$ for the height of the lowest body link at time t with the floor at $z = 0$:

- **Elevated support.** $z_{\min}$ holds above $0.20\,\mathrm{m}$ for at least $0.6\,\mathrm{s}$ (standing or climbing on a box).
- **Ground penetration.** $z_{\min}$ falls below $-0.12\,\mathrm{m}$, or below $-0.08\,\mathrm{m}$ for over 80% of the clip (legs driven through the floor by a faulty retarget).
- **Floating feet.** For at least $0.5\,\mathrm{s}$ *both* feet stay more than $0.02\,\mathrm{m}$ above the floor—measured as MuJoCo's exact geom-to-plane distance (`mj_geomDistance`) from the foot collision geometry, not a link-origin proxy—while the pelvis is upright and at standing height and the body is nearly stationary. A free-standing humanoid cannot hold both feet off the floor while standing still, so this is infeasible on flat ground whatever the cause (a low platform, a retarget float, or a slow rise; a ballistic jump is excluded by the stationary gate). The clean ground stance sits at $-0.004 \pm 0.004\,\mathrm{m}$, so the $0.02\,\mathrm{m}$ gate is several σ clear of the floor.

This pass removes a further 153 clips. All thresholds are tuned for the G1 retarget at $50\,\mathrm{Hz}$.

F Evaluation

F.1 Benchmark metrics

We evaluate SONIC and SuperMimic on the same filtered BONES-SEED training-set benchmark, containing 129,514 evaluated motions. For SONIC we evaluate its released checkpoint and its v1.1 checkpoint [1]. Each motion plays once from its first frame, with the same simulation for every policy: foot friction 1.0, no domain randomization or observation noise, and deterministic physics. Each policy uses its own actuator gains and action scale, and receives its observations as its training environment builds them. A motion is marked as failed if the rollout triggers SuperMimic's training termination (Appendix A). SONIC's own success criterion fails a motion on root and end-effector height deviations above $0.25\,\mathrm{m}$, without an orientation term [1].

F.2 Failed motions

Table A.8 counts failed motions per motion group. SONIC fails whole groups (all 44 flips, all 22 side rolls, and all 8 motions of one dance routine) and 15 of the 24 broad jumps. Its v1.1 checkpoint fails 239 motions, among them the same flips and dance motions and 19 of the 22 side rolls. After $3\,\mathrm{h}$ on one GPU, SuperMimic's failures concentrate in a few small groups of hard motions: the first five groups of Table A.8, 195 motions, hold 14 of its 25 failures. In no group does it fail more motions than SONIC. After $9.8\,\mathrm{h}$ on two GPUs, SuperMimic fails on none. Figure A.1 shows the count over training. The benchmark motions are training motions for both methods, so the counts measure how fully each policy covers these training motions, not generalization to new motions.

阶段	片段
原始 BONES-SEED（G1 重定向）	142，220
名称过滤器后	129,667
可行性过滤器后	129，514

表 A.7：**过滤前及经过各过滤器后的 BONES-SEED 片段数量**。

坐、攀爬、从 50 厘米高处跳下、越过围栏查看。此过程（改编自 SONIC 发布的数据处理代码中的关键词列表 [1], 并补充了其遗漏的若干术语）移除了12，553 个片段。

可行性过滤器。名称过滤器会漏掉那些名称未揭示问题的片段（例如走上箱子的常规动作）。我们通过 MuJoCo 正向运动学运行每个动作，并根据生成的身体连杆和足部几何轨迹进行标记，其中 $z_{\min}(t) = \min_b z_b(t)$ 表示时间 t 时最低身体连杆的高度（地面位于 $z = 0$）：

- **Elevated support**。$z_{\min}$ 在至少 0.6 秒内保持高于 0.20 米（站立或攀爬于箱子上）。
- **地面穿透**。$z_{\text{最小值}}$ 低于 $-0.12\mathrm{m}$，或在超过 80% 的片段中低于 $-0.08\mathrm{m}$，（腿部因重定向故障而穿入地面）。
- **脚部悬浮**。在至少 0.5 秒内，双脚均保持在离地 0.02m 以上——该距离通过 MuJoCo 精确计算的足部碰撞几何体到平面的距离（mj geomDistance）得出，而非连杆原点代理——同时骨盆保持直立且处于站立高度，身体近乎静止。一个自由站立的人形机器人无法在静止状态下使双脚同时离地，因此无论在平地上由何种原因导致（低平台、重漂移浮或缓慢上升；弹道跳跃已被静止门限排除），此情况均不可行。干净的站立姿态位于 $-0.004 \pm 0.004\mathrm{m}$，因此 0.02m 的门限与地面有数个 σ 的余量。

本次过滤额外移除了 153 个片段。所有阈值均针对 50 Hz 的 G1 重定向进行了调优。

F 评估

F.1 基准测试指标

我们在相同的经过过滤的 BONES-SEED 训练集基准测试上评估 SONIC 和 SuperMimic，该基准测试包含 129,514 个被评估的动作。对于 SONIC，我们评估其发布的检查点及其 v1.1 检查点 [1]。每个动作从其第一帧开始播放一次，所有策略使用相同的模拟设置：足部摩擦为 1.0，无域随机化或观测噪声，并采用确定性物理。每个策略使用其自身的执行器增益和动作尺度，并接收由其训练环境构建的观测。如果 rollout 触发了 SuperMimic 的训练终止（附录 A），则该动作被标记为失败动作。SONIC 自身的成功标准会在根节点和末端执行器高度偏差超过 $0.25\mathrm{m}$时判定动作失败，且不包含朝向项 [1]。

F.2 失败动作

表A.8 统计了每个动作组中的失败动作。SONIC 会导致整个动作组失败（全部 44 个空翻、全部 22 个侧滚，以及一个舞蹈动作中的全部 8 个动作），并且 24 个立定跳远中有 15 个失败。其 v1.1 检查点导致 239 个动作失败，其中包括相同的空翻和舞蹈动作，以及 22 个侧滚中的 19 个。在单个 GPU 上训练 3 小时后，SuperMimic 的失败集中在少数几个高难度动作的小组中：表 A.8 的前五个小组共包含 195 个动作，占据了其 25 个失败中的 14 个。在任何小组中，其失败动作数量均未超过 SONIC。在两个 GPU 上训练 9.8 小时后，SuperMimic 不再有任何失败。图 A.1 展示了训练过程中的失败数量变化。基准动作是两种方法的训练动作，因此这些计数衡量的是每种策略对这些训练动作的覆盖完整程度，而非对新动作的泛化能力。

Motion group (motions)	SONIC	SuperMimic 1 GPU × 3 h	SuperMimic 2 GPUs × 9.8 h
side rolls (22)	**22**	2	0
flips: 360°, 180°, 90° (44)	**44**	1	0
fire-in-the-hole turns (20)	3	3	0
crossed legs (10)	9	0	0
lying, resting and standing up (99)	28	8	0
broad jumps (24)	15	0	0
dance routine (8)	**8**	0	0
all other motions (129,287)	50	11	0
total (129,514)	179	25	0

Table A.8: **Failed motions per motion group on the filtered BONES-SEED training-set benchmark.** SONIC: released checkpoint. Bold: the policy fails every motion in the group.

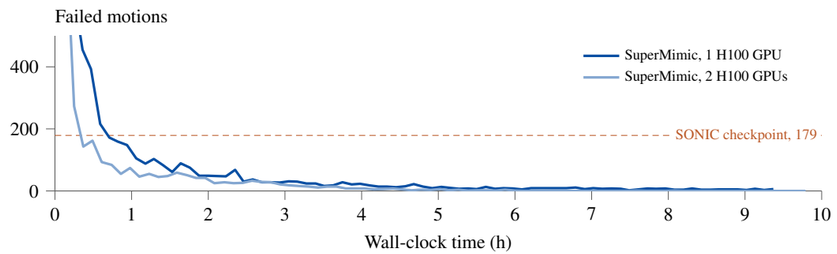


Figure A.1: **Failed motions during training on the filtered BONES-SEED training-set benchmark.** Checkpoints are scored every 4000 iterations, for the runs of Figure 3(a). SONIC: released checkpoint.

F.3 Experiment protocols

Figure 3. SuperMimic trains on the filtered BONES-SEED set (Appendix E). The dataset-length sweep in (b, c) uses subsets of 1, 7, 40, and 264 h.

G Velocity-command task

G.1 Training setup

The velocity encoder of §4.5 maps the command (v_x, v_y, ω_z) and a feature of the last 10 frames of $\boldsymbol{o}$ to the shared latent.

Each encoder is trained from scratch in its own run, jointly with the motion-tracking interface, for 100,000 iterations. Both runs use a walking and running library filtered from LAFAN1 and BONES-SEED. Half of the 4096 simulator environments and half of each imagined batch use the velocity interface, so $p(\mathcal{I})$ is uniform; unlike Algorithm A.1, which samples one interface per update, each imagined batch mixes both.

Commands. The command $\boldsymbol{g}_t = (v_x, v_y, \omega_z)$ stacks the target planar root velocity in the heading (yaw-only) frame and the target world-frame yaw rate. A reference command is the centered 25-frame mean of the reference root velocity in the reference's heading frame. For the RL encoder, each environment resamples its command source at every reset and every 250 control steps (5 s): with probability 0.5 it reads the command from the reference clip at every step, and otherwise it holds a command drawn uniformly from $v_x \in [-1.5, 2]\,\mathrm{m/s}$, $v_y \in [-1.32, 1.32]\,\mathrm{m/s}$ and $\omega_z \in [-3.35, 3.35]\,\mathrm{rad/s}$, zeroed (standing) with probability 0.1.

动作组（动作）	SONIC	SuperMimic 1 个 GPU × 3h	SuperMimic 2 个 GPU × 9.8h
侧滚（22）	**22**	2	0
空翻： 360°, 180°, 90° (44)	**44**	1	0
快速急转 (20)	3	3	0
交叉腿 (10)	9	0	0
躺卧、休息和站立 (99)	28	8	0
立定跳远 (24)	15	0	0
舞蹈动作 (8)	**8**	0	0
所有其他动作（129,287）	50	11	0
总计（129,514）	179	25	0

表 A.8：**过滤后的 BONES-SEED 训练集基准上每个动作组的失败动作**。SONIC：已发布的检查点。加粗：策略在该组中每个动作均失败。

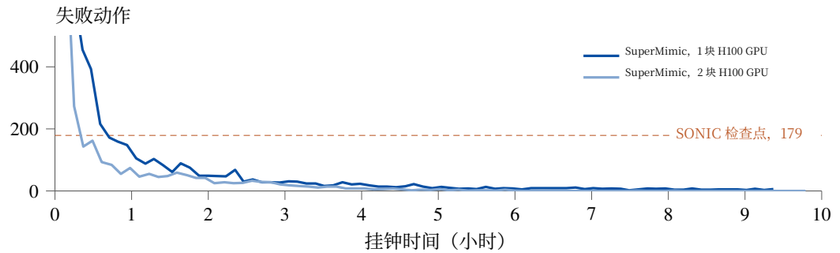


图 A.1：**在过滤后的 BONES-SEED 训练集基准上训练期间的失败动作**。检查点每4000 次迭代进行一次评分，针对图3(a) 中的运行。SONIC：已发布的检查点。

F.3 实验协议

图3。SuperMimic 在过滤后的 BONES-SEED 数据集上进行训练（附录 E）。(b, c) 中的数据集长度扫描使用了 1、7、40 和 264 小时的子集。

G 速度指令任务

G.1 训练设置

§ 4.5 中的速度编码器将指令 (v_x, v_y, ω_z) 以及过去 10 帧的$\boldsymbol{o}$ 特征映射到共享隐变量。

每个编码器都在其独立的运行中从头开始训练，并与运动追踪接口联合训练，持续 100,000次迭代。两次运行均使用从 LAFAN1 和 BONES-SEED 中筛选出的行走与跑步库。4096 个模拟器环境中有一半以及每个想象批次中的一半使用速度接口，因此 $p(\mathcal{I})$ 呈均匀分布；这与算法A.1不同，后者每次更新仅采样一个接口，而每个想象批次则混合使用两者。

指令。指令 $\boldsymbol{g}_t = (v_x, v_y, \omega_z)$ 将航向坐标系（仅偏航）中的目标平面根速度与目标世界坐标系偏航速率进行堆叠。参考指令是参考片段航向坐标系中参考根速度的居中 25 帧均值。对于强化学习编码器，每个环境在每次重置及每 250 控制步（5 秒）时重新采样其指令源：以 0.5 的概率在每一步从参考片段读取指令，否则则保持一个从 $v_x \in [-1.5, 2]\,\mathrm{m/s}$ $v_y \in [-1.32, 1.32]\,\mathrm{m/s}$、$\omega_z \in [-3.35, 3.35]\mathrm{rad/s}$ 中均匀抽取的指令，或以 0.1 的概率设为零（站立）。

Encoder. On rollouts with reference commands, a mean-squared loss of weight 0.1 pulls the velocity latent toward the motion encoder's latent at the same step, with a stop-gradient on the motion latent.

Rollouts, reward, and termination. Each interface's imagined rollouts start from states logged by its own simulator environments. Velocity rollouts of the RL encoder use the velocity-tracking reward and a fall-only termination (Table A.6); those of the distillation baseline use the motion-tracking reward and termination of Appendix A.

G.2 Step test

Protocol. From standing, the policy receives a fixed command for $5\,\mathrm{s}$; each command is tested in four trials. The relative steady-state error is the norm of the mean velocity error over the last $2\,\mathrm{s}$, divided by the norm of the command, with the root velocity measured in the heading frame. The per-command value is the median over the trials that did not fall. The 672 step responses form a 15×15 grid on each of three planes, spanning the uniform command range above, with the standing command left out. In Figure 7, errors above 0.9 are drawn at 0.9, and a policy that fell in all four trials is drawn at the cap.

Results. The RL encoder has the lower error on 650 of the 672 step responses, and the distillation encoder on 15. Their median relative errors are 0.16 and 0.49. Distillation falls in every trial on seven commands, all combining translation with a turn of at least $2.8\,\mathrm{rad/s}$, where the motion library has almost no data. Each policy has one training seed.

编码器。在使用参考指令的 rollout 中，权重为 0.1 的均方损失会将速度隐变量拉向同一时刻运动编码器的隐变量，并对动作隐变量执行梯度停止。

rollout、奖励和终止条件。每个接口的想象 rollout 均从其自身模拟器环境记录的状态开始。RL 编码器的速度 rollout 采用速度跟踪奖励和仅跌倒终止（表A.6）；蒸馏基线的 rollout 则采用运动跟踪奖励以及附录A中定义的终止条件。

G.2 步态测试

协议。从站立状态开始，策略接收一个持续5s的固定指令；每条指令进行四次试验。相对稳态误差定义为最后2s内平均速度误差的范数除以指令的范数，其中根速度在航向坐标系中测量。每条指令的取值为未发生摔倒的试验结果的中位数。672 个阶跃响应在上述均匀指令范围内（不包括站立指令）于三个平面上各形成一个网格。在图7中，大于 0.9 的误差按 0.9 绘制，而在四次试验中均摔倒的策略则按上限值绘制。

结果。在 672 个阶跃响应中，强化学习编码器在 650 个上的误差较低，而蒸馏编码器在 15 个上较低。它们的中位数相对误差分别为 0.16 和 0.49。在七条指令的每次试验中，蒸馏均出现摔倒，这些指令均结合了平移与至少 $2.8\mathrm{rad/s}$ 的转向，而动作库中几乎没有任何相关数据。每个策略均使用一个训练种子。